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My Health @ EU
eHealth Digital Service Infrastructure
Electronic cross-border health services in the EU

NCPeH Architecture Specifications

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ABSTRACT
This document defines the architectural view on the "NCP-In-A-Transparent-Box", describing the functionalities which are common and the "common components" implementing these functionalities. It documents which functionality can be developed in common and which "common components" will be specified.

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TABLE OF CONTENTS

TABLE OF CONTENTS 3

1 Executive Summary 6
1.1 Scope of this document 6
1.2 Glossary 6

2 Requirements and Security Considerations 6
2.1 Requirements 6
2.2 Security Considerations 7
2.3 Guidelines from eHealth DSI Solution Provider 7

3 NCP-In-A-Transparent-Box 8
3.1 Overview 8

Figure 1: NCP-in-a-Transparent-Box 8
3.2 Common Components 9

Figure 2: Common Components 9
3.2.1 Client and server 10
3.2.2 Security module 10
3.2.3 Transformation module 10
3.2.4 Terminology Services Access Manager 10
3.2.5 Audit module 10
3.2.6 ConfigurationAndMonitoringManager 11
3.2.7 NationalConnector 11
3.3 eHealth DSI Front-End Services Transactions 11
3.3.1 eHealth DSI Patient Identification Service 11

Figure 3: Patient Identification 12
3.3.2 eHealth EUDI Wallet Patient Identification 13
3.3.3 eHealth DSI Patient document services 15

Figure 4: List patient documents 15

Figure 5: Retrieve patient document 16
3.3.4 eHealth DSI Dispensation Service 17

Figure 6: Dispensation and discard 17
3.3.5 eHealth DSI Patient Identification Service (FHIR) 18
3.3.6 eHealth DSI Document Reference Service (FHIR) 19

Figure 8: Document Reference Search (FHIR) 19

Figure 9: Document Reference Search (FHIR) detail 20

Figure 10: Query bundle (FHIR) 21

Figure 11: Post DocumentReference (FHIR) 22
3.4 Nation-Specific NCP Components 22
3.5 Central Common Services Components 23
3.6 NCP-B Internet Front-end 23
3.6.1 System Overview 23

Figure 17: Front-End System Overview 24
3.6.2 Security 26

Figure 18: Security of Front-End solution (portal) 26

Figure 19: Session management in country B 26
3.6.3 HP Authentication 26
3.6.3.1 Use Case 27

Figure 20: SAML Web SSO Sequence Diagram 27

3.6.3.2 NCP Web-Portal SAML profile 28
3.6.3.3 SAML Authentication Context Classes 28
3.6.3.4 Metadata Distribution 28
3.6.4 Site Collection 28
Figure 21: Site diagram (Site Collection) 29
3.6.5 Application Life-Cycle 29
Figure 22: Application Lifecycle (Informative) 29
3.6.6 Page Life-Cycle 29
Figure 23: Page Lifecycle (NCP request handling) 30
3.6.7 HP Session Life-Cycle 30
Figure 20: HP Session Lifecycle 30
Figure 21: Example from NCP assertion trust 31
3.6.8 HP Patient (TRC) Session Life-Cycle 31
Figure 22: HP Patient (TRC) Session Lifecycle 31
4 Example Deployment Composition 31
Figure 23: Logical View for both NCPs (Note, the protocol terminators are removed, this picture needs updating) 32
4.1 Navigation Paths 32
4.1.1 NCP-B Role 32
Figure 24: Overview for NCP-In-A-Transparent-Box in NCP-B Role 33
4.1.2 NCP-A Role 33
Figure 25: Overview for NCP-In-A-Transparent-Box in NCP-A Role 34
4.1.3 Central Common Service Role 34
4.2 Deployment 34
Figure 26: Exemplary Deployment of the NCP Components 34
4.3 Implementation Platform 35
4.4 Security Setup for NCP 35
4.4.1 Security Zones 35
Figure 27: General eHealth DSI Trust Zones 36
4.4.2 NCP gateway Application 38
Figure 28: NCP Application Security 38
5 Components Design 38
5.1 Services 39
5.1.1 FHIR implementation 39
5.1.2 eHealth DSI Patient Identification Service 39
Figure 30: eHealth DSI Patient Identification Service Interface 39
5.1.3 eHealth DSI Patient Service 40
Figure 31: eHealth DSI Patient Service Interface 41
5.1.4 eHealth DSI Order Service 42
Figure 32: eHealth DSI Order Service Interface 42
5.1.5 eHealth DSI Dispensation Service 44
Figure 33: eHealth DSI Dispensation Service Interface 44
5.1.6 eHealth DSI OrCD Service 46
Figure 34 eHealth DSI OrCD Service Interface 47
5.1.7 eHealth DSI Patient Identification Service (FHIR) 48
5.1.8 eHealth DSI Document Reference Service (FHIR) 48

5.1.9 Anomaly Detection 49
5.2 Audit Component *49*
Figure 38: AuditTrailWriter component **50**
5.3 Audit Repository *50*
Figure 39: AuditRepository component **50**
5.4 Security Module *51*
Figure 40: SecurityManager component **51**
5.5 Translations and mappings services *52*
5.6 Terminology Services Access Manager *55*
Figure 42: TerminologyServiceAccessManager component **55**
5.6.1 Interaction Uses 57
Figure 43: Document Consumer WorkflowManager – TransformationManager Interaction Use (IN_WM_TM_IU) **58**
Figure 44: Document Provider WorkflowManager – TransformationManager Interaction Use (OUT_WM_TM_IU) **58**
5.7 NationalConnector *59*
Figure 45: NationalConnector component **59**

1 Executive Summary

1.1 Scope of this document

This document defines the architectural view on the "NCP-In-A-Transparent-Box", describing the functionalities which are common and the "common components" implementing these functionalities. It documents which functionality can be developed in common and which "common components" will be specified.
The "Box" is defined "Transparent" because every country can decide if it is worth for it to adopt all or only part of the components defined as potentially common. Furthermore, it describes on a high level the national part of the "NCP-In-A-Transparent-Box", namely the "NationalConnector".
To provide a light integration solution for those countries that at this moment do not plan a full integration the NCP Country-B functionality with their national infrastructure, the document also describes a Country B Front-end solution.
Certain functionality required by NCPs (e.g., distribution of configuration-relevant data) is to be implemented as central services and referred to as Central Common Services. The specification for the Central Common Services is not included in this document. However, the way in which the NCP will be connected to them is defined, together with the NCP component to perform the interconnection.

1.2 Glossary

[eHealth DSI Glossary].

2 Requirements and Security Considerations

For all NCP implementations, the requirements of the current eHealth documents are mandatory. These are not repeated in this document.

2.1 Requirements

The requirements of this section form the basis for the work of the eHealth DSI Solution Provider and Technical Community.

- eHealth DSI basic pillars must be respected and safeguarded
  - eHealth DSI will provide solutions that are validated for Cross-border interoperability of eP and PS
  - eHealth DSI eP and PS Cross-border interoperability must be based on a Legal and Regulatory framework which includes the signature of contractual agreements among the countries to state the legal responsibilities and assure the adequate level of truster
  - eHealth DSI eP and PS Cross-border interoperability must be based on already existing National eP and PS services
  - eHealth DSI eP and PS Cross-border interoperability must not interfere with National eP and PS services or request modification to them
  - eHealth DSI eP and PS Cross-border interoperability services must not decrease the level of security provided by every country to its Citizens
  - eHealth DSI will not develop European eP / PS services, but make existing national services accessible from all participating nations
- [NCPeH Components Specification] is based on the legal requirements, the functional requirements, the technical requirements,
- NCP in a transparent box means that every component has a defined interface and if needed can be easily substituted by a corresponding national implementation. It is agreed upon that the NCP must be provided as a very flexible framework.
- As many components as possible, adopting reasonable criteria, should be developed in common. However, this will not be possible for all functionalities of

the NCP.

- The possible strategies that might be followed to develop in common the NCP components and the Country-B Front-end will be described in the [NCPeH Components Specification], however it is the responsibility of eHOMB to choose among alternatives and adopt the selected one.

2.2 Security Considerations

eHealth DSI Multilateral Agreement document has to assure the respect of the Security Requirements, regardless the fact the NCP is acting as the data controller or the data processor[1].

All development has to be in line with all security requirements and the policies of all beneficiaries.

Making reference to ISO/IEC 27002, generally the main objectives of computer security can be summarized as:

- **Authenticity**: the identity of an actor has been proven as true;
- **Confidentiality**: information is accessible only to authorized users/[actors];
- **Integrity**: accuracy and completeness of information and processing methods;
- **Availability**: authorized users have access to information and associated assets when required;
- **Accountability/Non Repudiation (Liability)**: each communication and each data transaction can be tracked back to a certain originator in a traceable chain of activities.

In eHealth DSI system, the core of data processing of the eHealth DSI NCP, is related to process Health Care data. This highly sensitive data processing imposes that the: authenticity, confidentiality, integrity, availability and liability security objectives must be guaranteed through the suitable security requirements.

2.3 Guidelines from eHealth DSI Solution Provider

To enable the development of the "NCP-In-A-Transparent_Box" additional requirements have been defined by epSOS and now by eHealth DSI Solution Provider. These additional requirements are listed below:

- The NCP Gateway is composed of a common (eHealth DSI) part and an individual (nation specific) part. The components of the common part are specified in detail. The nation specific part is implemented in the "NationalConnector" that is not specified in detail but implements a stable interface that is defined in this document.
- The components in the common part might or might not be adopted by the different countries, according to their National Infrastructure and the country defined implementation policies.
- The architecture and the eHealth DSI transactions must be able to transport pseudonymised data that was sent by country A in a way that only pseudonymised data can be seen by NCP-B.
- The individual security requirements of each country, derived from [Security Services Specification], are documented in this document in Section 2.2. Organizational and technical means must be defined in the deliverable to make sure security requirements are fulfilled. If security requirements cannot be fulfilled, this must be documented.
- The authentication of the Healthcare Professionals (HP) is a prerequisite for any eHealth DSI transaction (usually Identification Handshake). The authentication of the HP is done by the identity provider which is part of the national infrastructure and must be separated from the NCP.

[1]Data Processor and Data Controller are defined in “Regulation (EU) 2016/679, Article 4 – Definitions –“.

- The identification request is checked in the common part of the NCP A in the way that the XML containing the patient traits is schema validated; the semantics of the traits are checked in the national part.
- The security audit that forms the basis for administrative tasks and reporting must be separated from the functional logging.

3 NCP-In-A-Transparent-Box

This section presents an overview of an "NCP-In-A-Transparent-Box". First it presents the components of the NCP. Then interactions between the components are presented.

3.1 Overview

The [System Architecture Specification] document describes in the Information System View the structure and the high level behaviour of the NCP gateway.
The NCP-In-A-Transparent-Box is one scenario of an NCP implementation that includes special requirements (see Chapter 2). Therefore, the composition of the NCP is slightly different from the architecture defined in the [System Architecture Specification]. The main difference is that the components are separated into "common components" and "nation-specific components" with the nation-specific components being encapsulated by the "NationalConnector".
To provide a stable basis for the common components, the interface of the NationalConnector must be defined. The NationalConnector is implemented as a black box having its subcomponents hidden from the NCP. They are nevertheless part of the NCP because they implement core functionality of the NCP and fulfil functional and non-functional requirements.

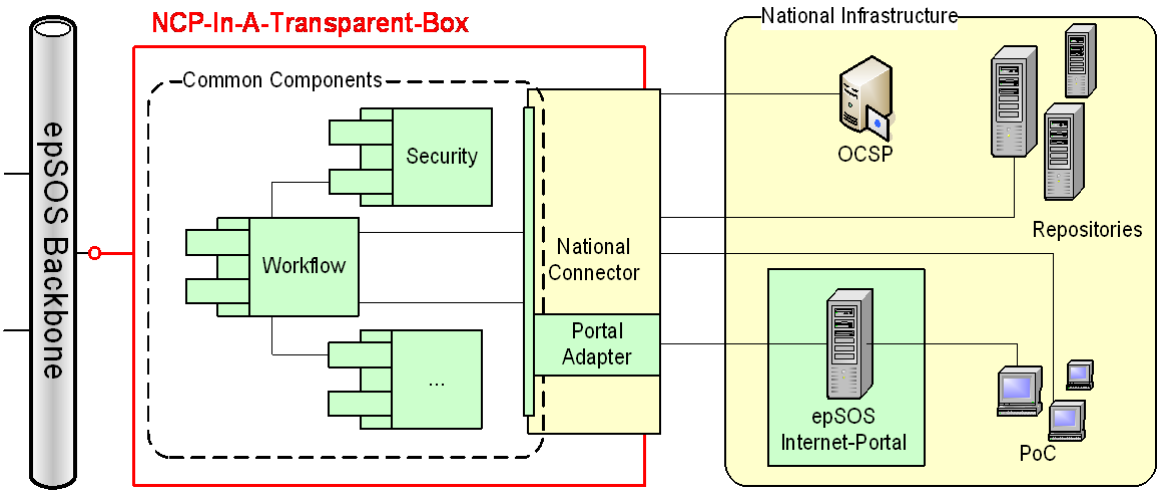


Figure 1: NCP-in-a-Transparent-Box

The NCP-In-A-Transparent box is meant to be a flexible framework of common components that may be used by country to fulfil the requirements of their pilot sites and eHealth DSI regarding the eHealth DSI transactions. Figure 1 shows by colour which components MAY be developed in common (green) and which are specific for each country.
Each country is responsible for its own and unique implementation of the NCP gateway. We assume that for reducing the costs, each country will try to use as many common components as possible. Nevertheless a reasonable part of the NCP gateway remains nation specific and therefore cannot be specified nor jointly developed by eHealth DSI. The focus of this chapter lays on the common components that are defined in section 3.2.
Common central services are not shown in that abstract figure, but they will be connected to the eHealth DSI Backbone.

The national connector encapsulates the nation-specific part of the NCP hence it is not a common component. Nevertheless it has an interface that is defined in common to achieve interoperability between the common components and the national part.

3.2 Common Components

The following components have been identified by the eHealth DSI Solution Provider and Technical Community:

- ClientConnector
- Client
- Server
- SMP
- Security module
- Transformation module
- TerminologyServiceAccessManager
- Audit module
- ConfigurationAndMonitoringManager
- NationalConnector

The following components were discussed but have been discarded

- Time synchronisation is done by the platform; therefore there will be no dedicated component for that task.
- A LoggingManager is not needed since the logging will be processed by a dedicated logging framework. The logging that is needed for reporting and administrative tasks has to be separated from the security audit.
- There will be a management console that is used for administrative tasks, for monitoring and for reporting. The management console will be able to read the logfiles and will be able to provide the configuration data.

The definition of the components is very close to the basic architecture presented in [System Architecture Specification]. On the one hand, the architecture of the NCP-In-A-Transparent-Box should allow having as much of commonly developed components as possible. On the other hand, the architecture should be flexible enough for supporting the variety of national solutions implemented by countries. An overview of the components is given in Figure 2 Common Components.

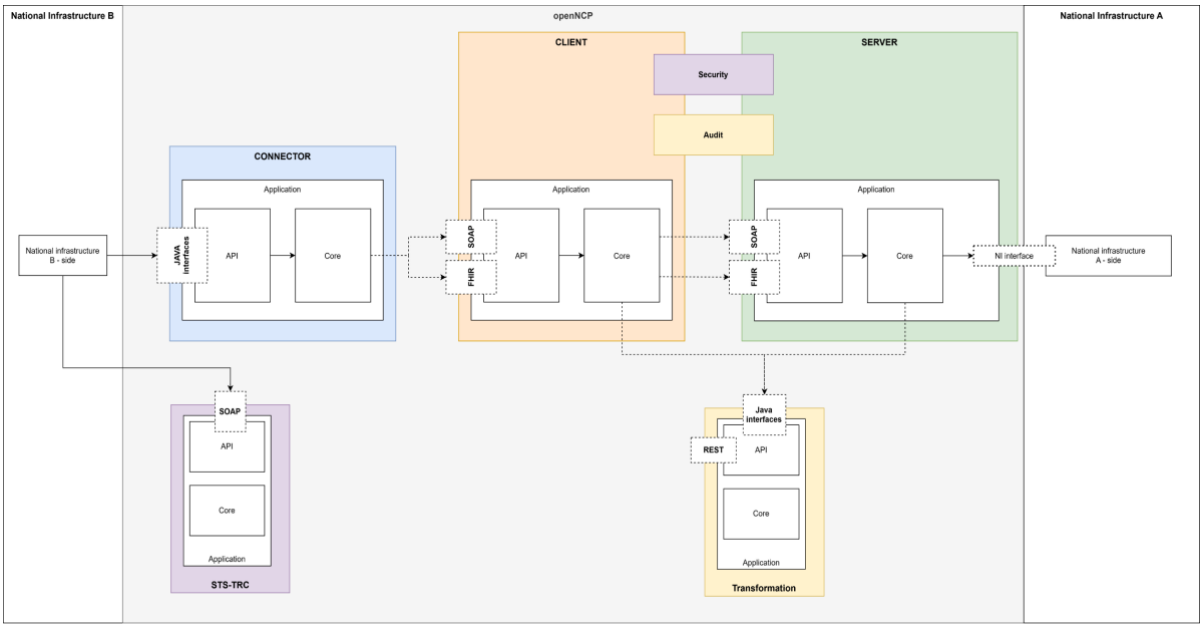


Figure 2: Common Components

Components that are not common but part of the NCP gateway are grouped in the NationalConnector.

The basic concepts of the architecture are derived by the logical sectioning of the NCP gateway into three sections (vertical layers). The workflow manager is the component which is responsible for managing communication between the layers.

3.2.1 Client and server

The eHealth DSI client and server application is organized into three layers (exposed as modules) that together deliver the total NCP functionality. The client acts as the NCP-B side application while the server acts as the NCP-A side application. Both exposes the SOAP-based and HL7 FHIR interfaces, enforces all required security protocols (TLS, SAML, JWT, and others), and orchestrates the complete suite of NCP operations in accordance with the defined business rules.

The application layer handles service bootstrap and configuration. It contains the main entry point along with environment profiles, dependency-injection settings, server ports, and any other high-level parameters needed to initialize and launch the client. The API layer then takes over all transport- and protocol-level concerns: it routes incoming requests, parses and validates payloads, applies security interceptors for authentication and authorization and registers the appropriate FHIR resource providers.
Finally, the core layer encapsulates the domain logic itself: it implements all business workflows, data transformations, auditing and inter-service coordination required by the NCP specification, ensuring that every operation complies with both technical and regulatory requirements.

3.2.2 Security module

The Security module is used for certificate validation and XML-Signature creation and validation and handles assertions wrapped in JWT tokens for FHIR. It is mandatory that the security has a list of all trusted certificates to check whether the given certificate ismember of the circle of trust. The certificate validation includes the mathematical check, the check of the validity in time and the OCSP call.

3.2.3 Transformation module

This component will be called in two different scenarios: either for data transformation from a national language to the eHealth DSI Reference Terminology or for data transformation from the eHealth DSI Reference Terminology to a national language.

3.2.4 Terminology Services Access Manager

This component is called by TransformationManager. The component is responsible for translating a given concept designation into the requested target language as well as transcoding a given "local" coded concept into the appropriate eHealth DSI coded concept using the information present in the Terminology Repository. The Terminology Repository is a database that is a part of the NCP and represents the eHealth DSI Reference Terminology. The content of the Terminology Repository is specific for each country. It is the responsibility of the country to maintain and update the content of the Terminology Repository.

3.2.5 Audit module

As required every transaction in eHealth DSI must be audited with very limited information. National requirements for an extended auditing must be realized in the NationalConnector or the national infrastructure.

In future versions it may be an option to start the auditing in an asynchronous manner. For the current version this option will be neglected due to simplicity.

Note, that it should be differentiated between logging and audit. Audit captures business-level events (e.g., request for PS) while logging occurs on a lower level and is responsible for capturing application-level events.

3.2.6 ConfigurationAndMonitoringManager

The ConfigurationAndMonitoringManager provides a monitoring console that is used for managing configurations of the nation-specific components as well as monitoring their status.

3.2.7 NationalConnector

The NationalConnector is not a common component, but a collection of adapters to the national protocols and data formats. The NationalConnector makes use of a common exposed API as required by the WorkflowManager as well as exposes a common API to the WorkflowManager itself. The interface to the national infrastructure, however, is country-specific with no restrictions imposed on it.

3.3 eHealth DSI Front-End Services Transactions

This section describes the end2end interactions between the components of the NCP that correspond to invocations of the eHealth DSI front-end services:

- eHealth DSI Patient Identification Service
- eHealth DSI Patient Service
- eHealth DSI Order Service
- eHealth DSI Dispensation Service
- eHealth DSI OrCD Service
- eHealth DSI Patient Identification Service (FHIR)
- eHealth DSI Document Reference Search (FHIR)

The sequence diagrams provided below shows in which order and by which method calls the NCP components are invoked when one of the front-end services is called.

3.3.1 eHealth DSI Patient Identification Service

There are different alternatives to discover a Patient ID and therefore different attributes to be used by calling NCP-B. But all of them result in providing NCP-A with a PID (Patient Identity).

Figure 3 depicts the transaction as a sequence diagram, where each call has been distilled to its core structure and its name streamlined to convey only its primary purpose:

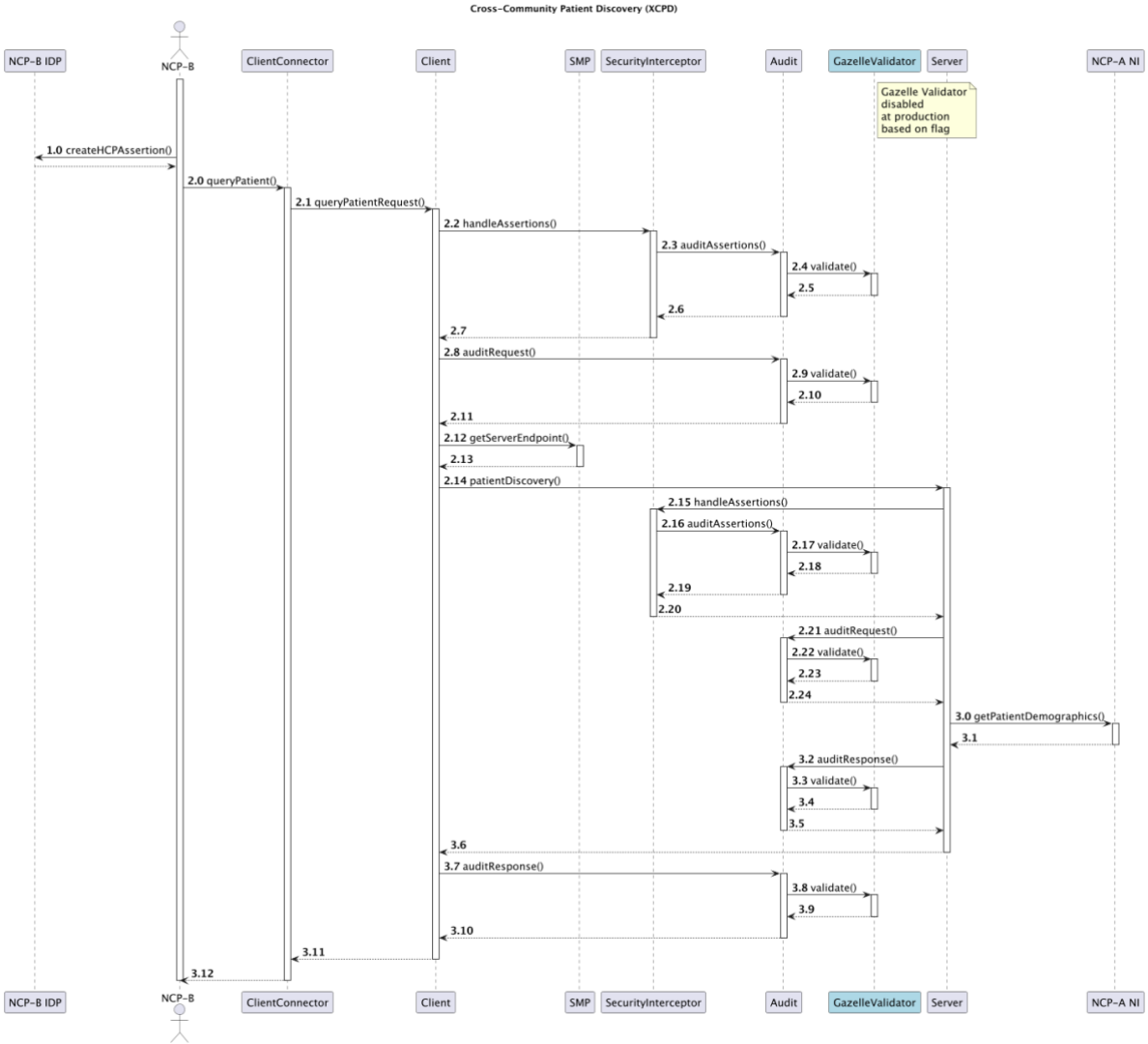
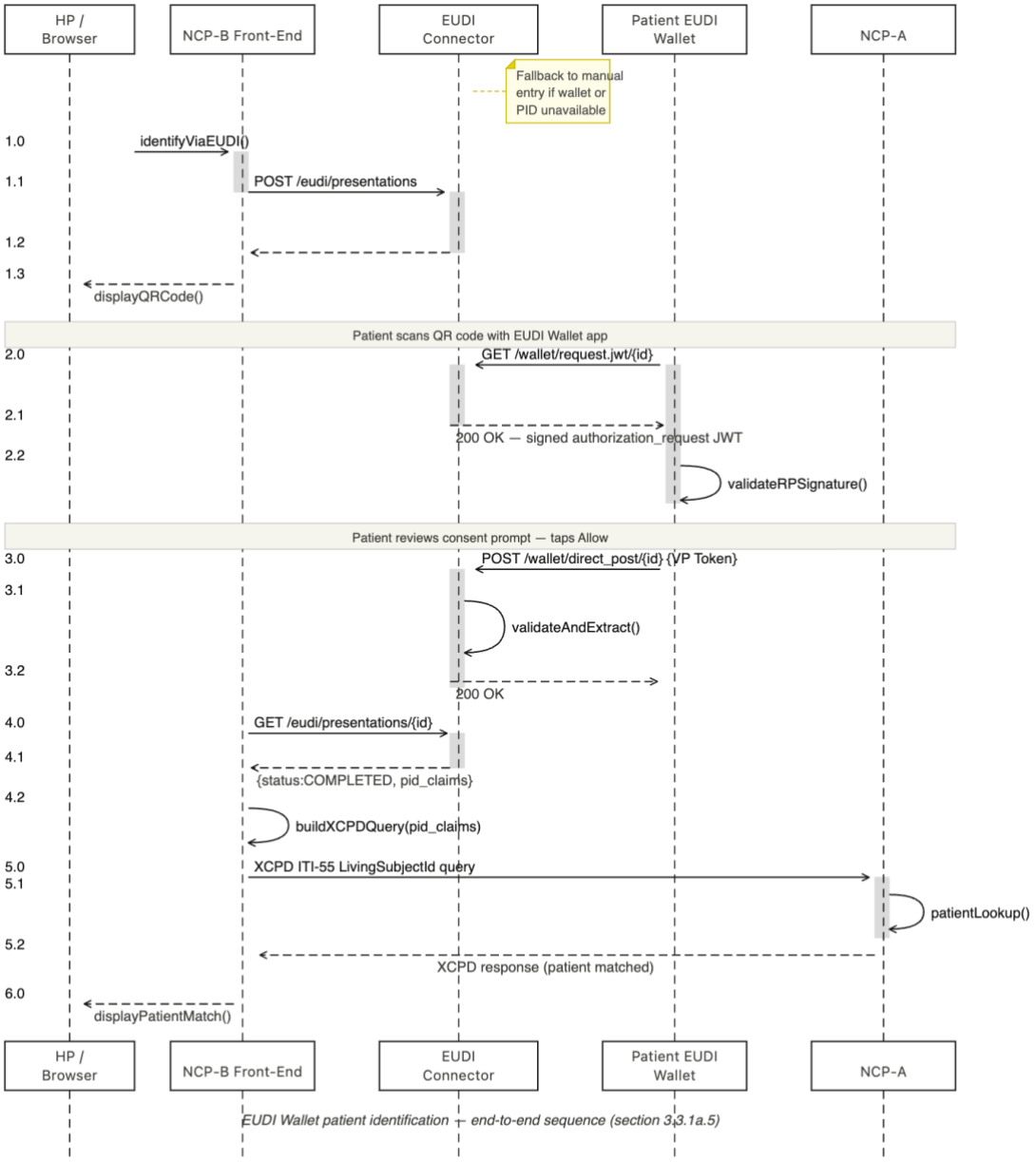


Figure 3: Patient Identification

3.3.2 eHealth EUDI Wallet Patient Identification

The EUDI integration is implemented as a dedicated connector module within the NCP-B deployment. This module acts as an OpenID4VP Relying Party and exposes two internal REST endpoints consumed by the NCP-B Front-End, and two public HTTPS endpoints consumed by the patient's EUDI Wallet app.



Since the *personal_administrative_number* field is populated at member state discretion. If absent, the NCP-B should fall back to standard demographic-based XCPD or PDQm.
The /wallet/request.jwt/{id} and /wallet/direct_post/{id} endpoints MUST be reachable from the public internet (the patient's mobile device is on a mobile network).

The EUDI-based patient identification event MUST be audited in the eHealth DSI audit trail in accordance with the existing ATNA audit requirements. The audit record MUST indicate that patient identification was performed via EUDI Wallet

presentation, including the issuing member state of the PID credential (derived from the issuer certificate). The full VP Token MUST NOT be stored in the audit trail.

In production deployments, the EUDI component MUST be registered as an EUDI Wallet Relying Party with the competent authority of the member state in which it operates, in accordance with Article 5b of the eIDAS 2.0 Regulation. The RP registration certificate MUST be used to sign the OpenID4VP authorization_request JWT, enabling the patient's wallet to display a verified relying party name during the consent prompt.

Fallbacks:

- If the patient does not have an EUDI Wallet app, or declines consent, the HP MUST be able to fall back to the existing manual patient identification flow.
- If the personal_administrative_number claim is not present in the presented PID, the connector MUST fall back to a name/DOB-based XCPD query.
- If the EUDI Connector callback endpoint is unreachable (e.g. network timeout), the front-end MUST surface a clear error and offer the manual identification alternative.

3.3.3 eHealth DSI Patient document services

The transaction is visualized by sequence diagrams displayed at Figure 4 and Figure 5:

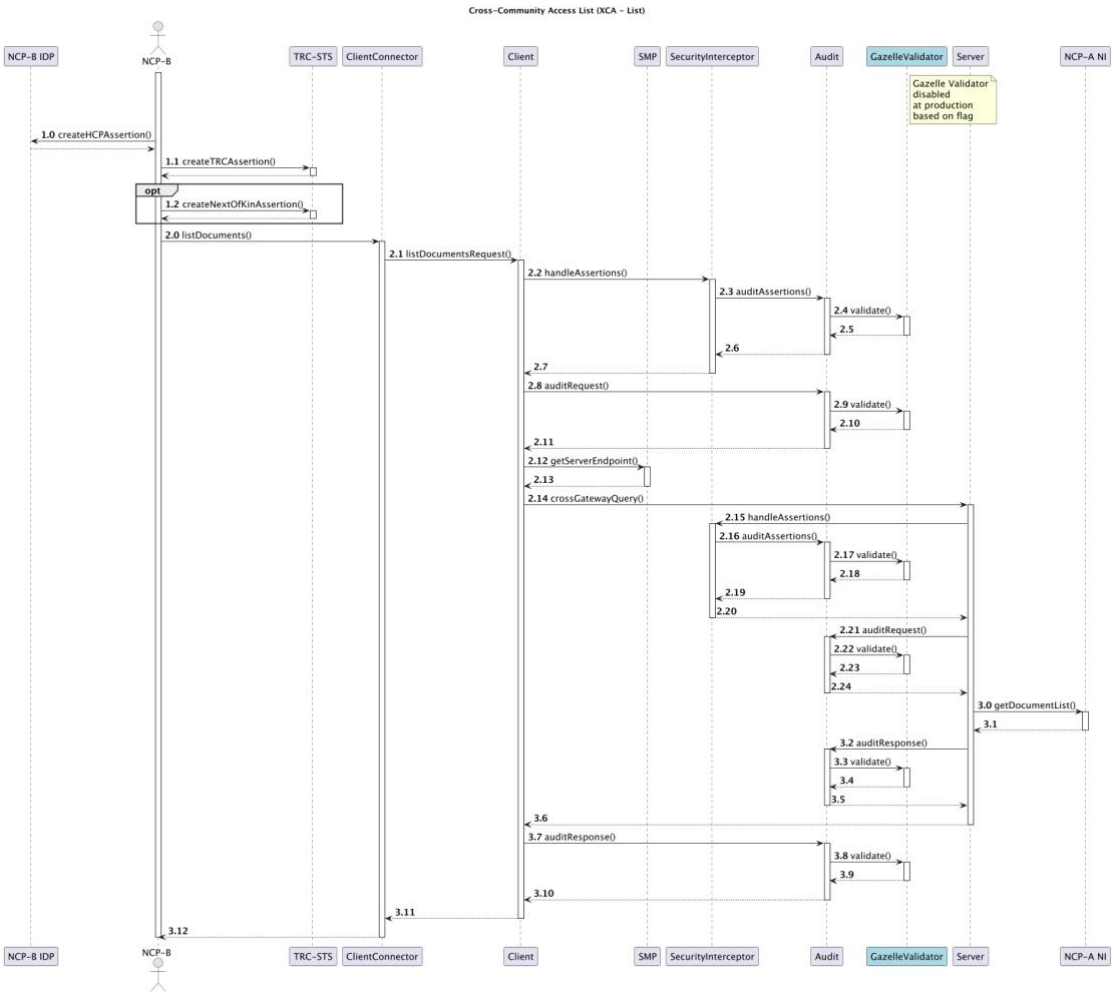


Figure 4: List patient documents

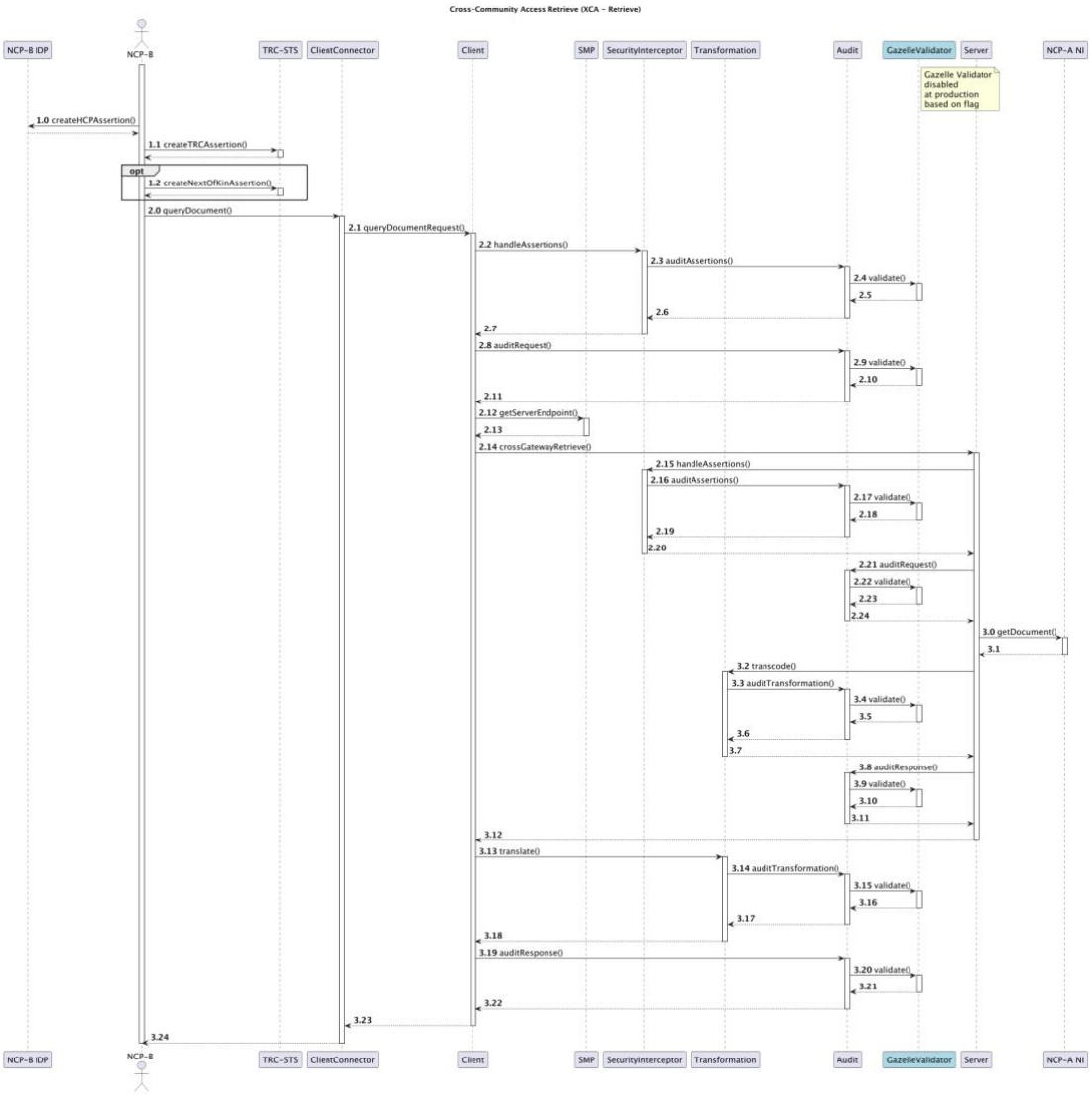


Figure 5: Retrieve patient document

3.3.4 eHealth DSI Dispensation Service

eHealth DSI should enable an HP to dispense a medication or revoke an already provided dispensation. Here the sequence diagram reflecting the dispense / discard calls of the NCP's Dispensation Service is provided

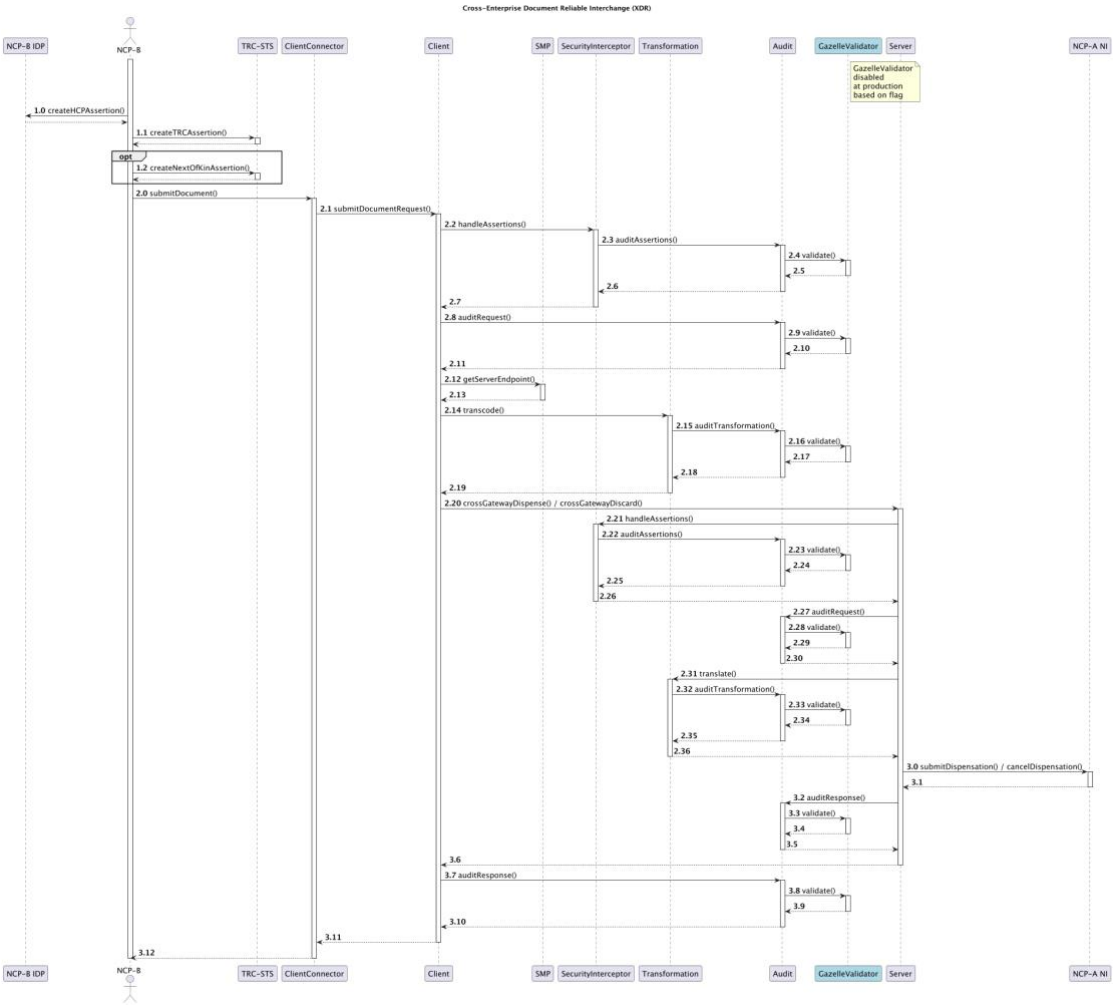


Figure 6: Dispensation and discard

3.3.5 eHealth DSI Patient Identification Service (FHIR)

The FHIR patient identification service allows to search for patients using the following parameters:

- businessIdentifier (optional)
- name (optional)
- birthdate (optional)
- gender (optional)

The way this is implemented in OpenNCP can visualized in the following sequence diagram:

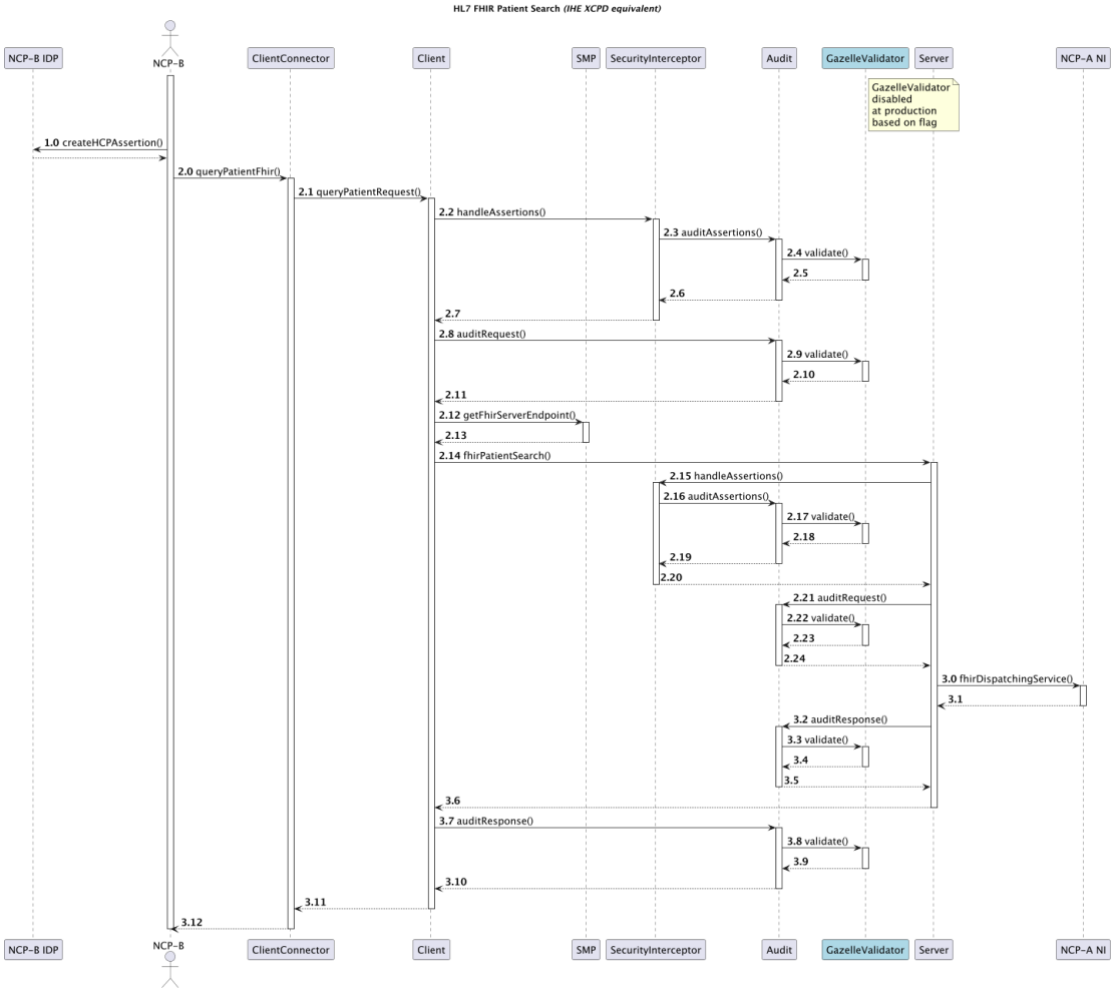


Figure 7 : Patient Identification (FHIR)

3.3.6 eHealth DSI Document Reference Service (FHIR)

The FHIR Document Reference service allows to retrieve lpatient data for a given patient in another country,retrieve relevant information from the country and send information to that country. The information is exchanged in the form of documents. These documents represent laboratory results, medical imaging reports & manifests, hospital discharge reports and vaccination card reports as supported by the eHN prioritised services (Laboratory Results, Medical Imaging, Hospital Discharge Report, Vaccination History).

Thedocument reference search allows the following parameters:

- Document type (mandatory)
- Patient Business Identifier (mandatory)
- Date (mandatory)
- Type of content (optional)
- Study type (optional)

The way this is implemented in OpenNCP can be seen in the following sequence diagrams:

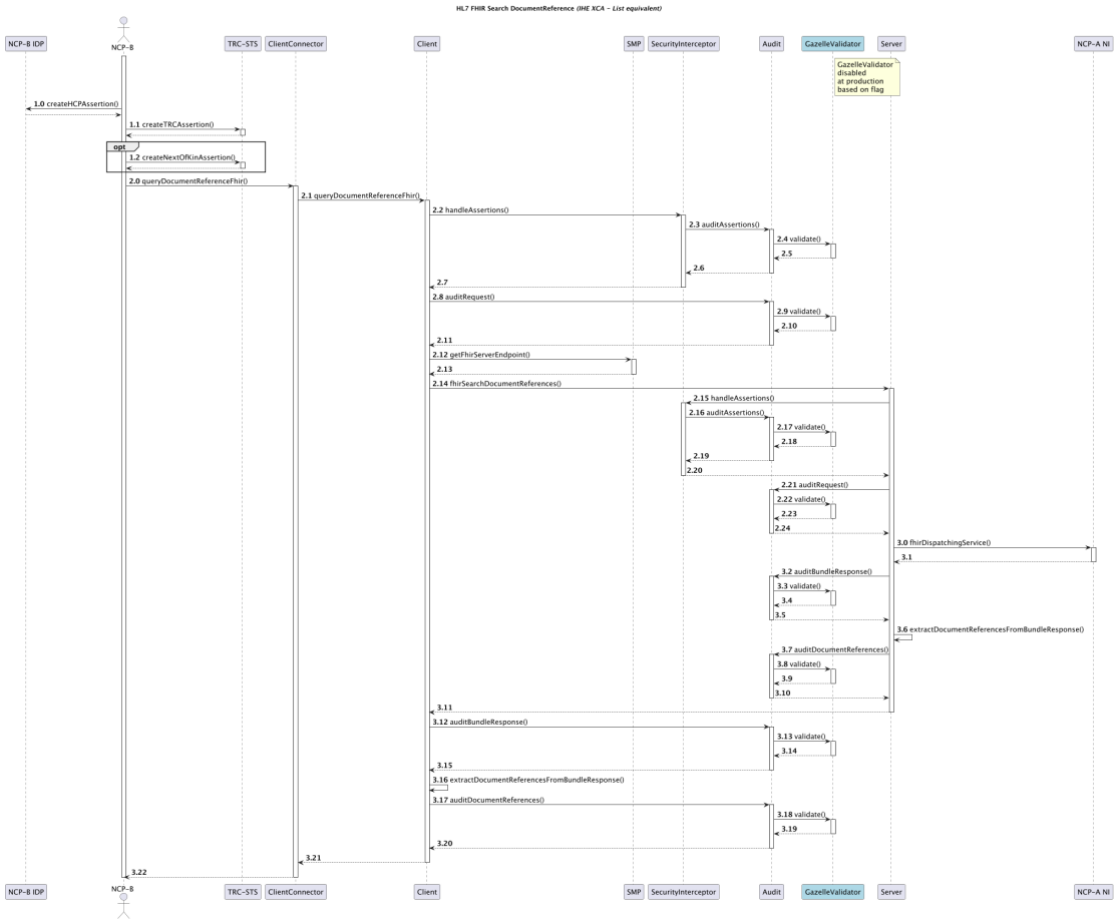


Figure 8: Document Reference Search (FHIR)

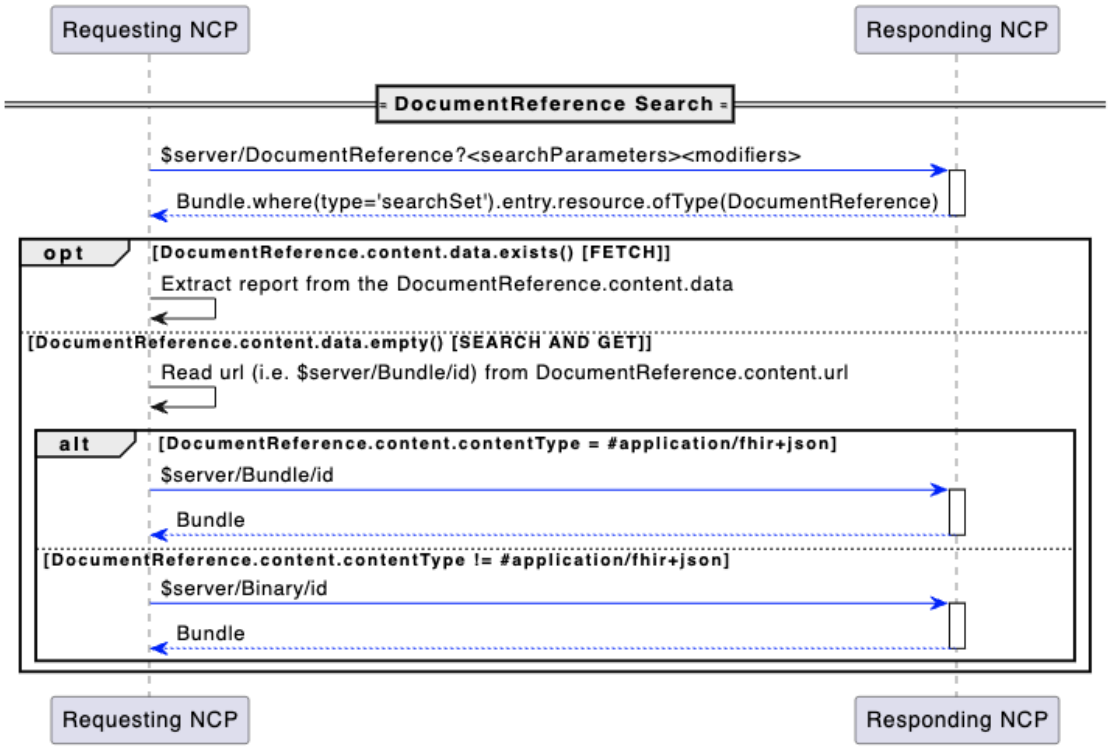


Figure 9: Document Reference Search (FHIR) detail

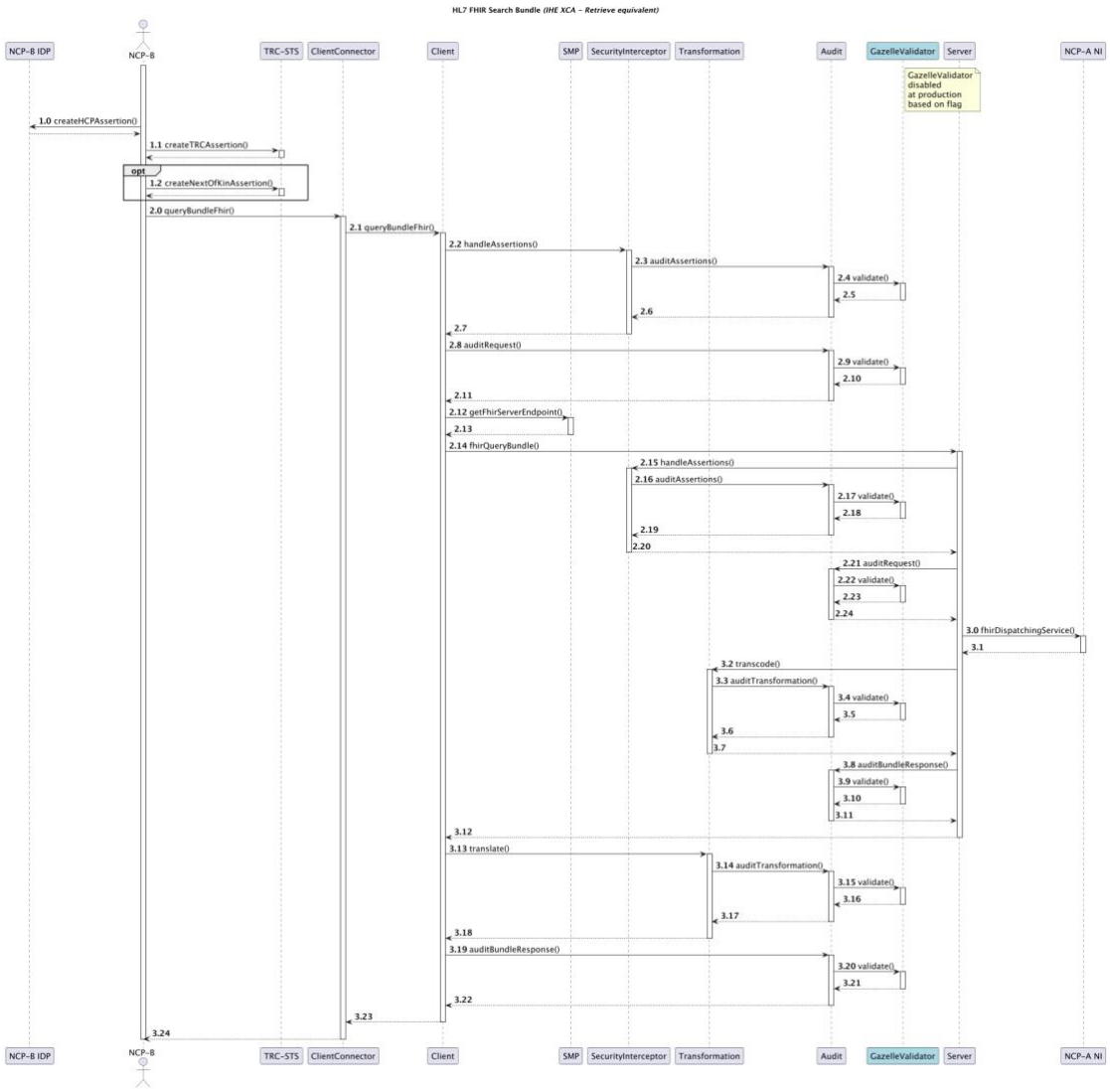


Figure 10: Query bundle (FHIR)

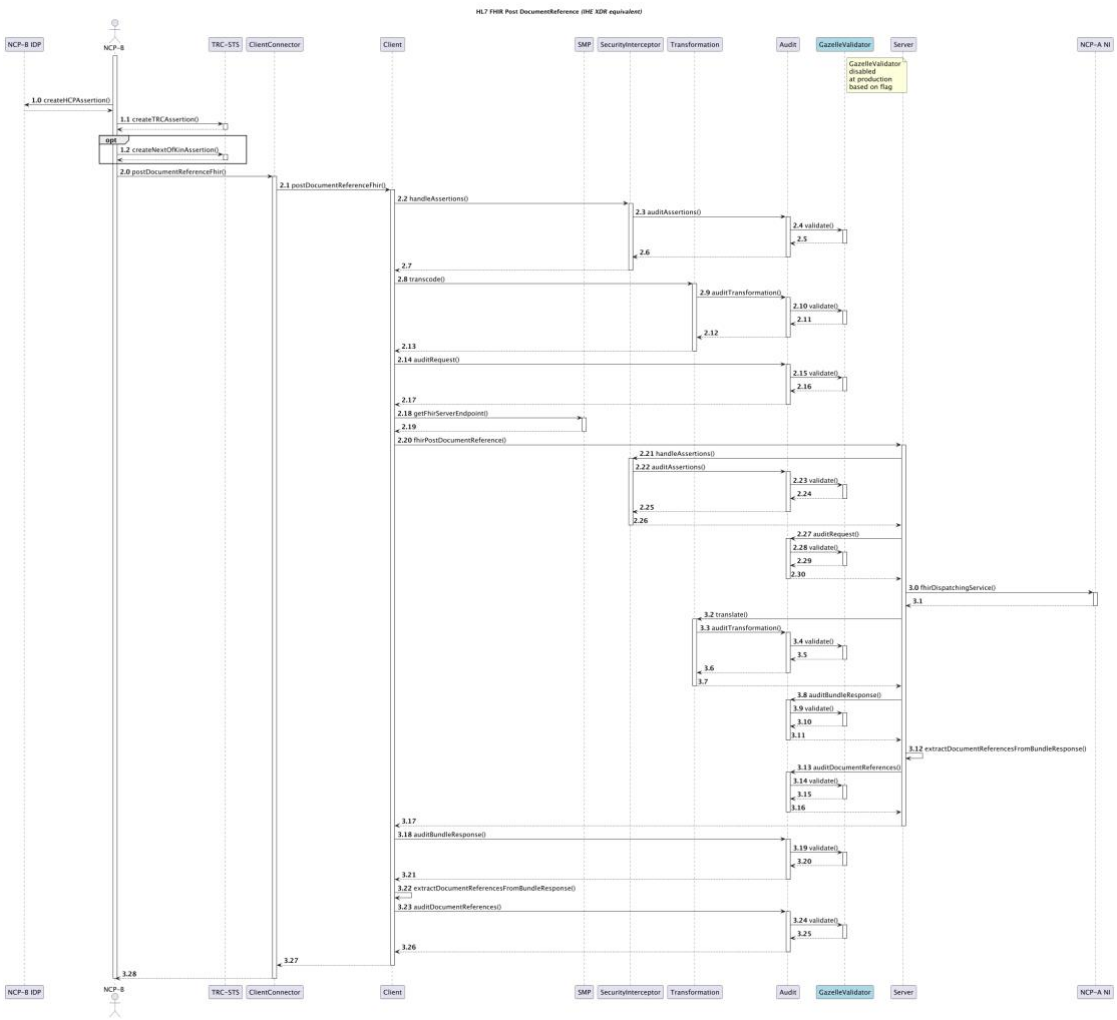


Figure 11: Post DocumentReference (FHIR)

3.4 Nation-Specific NCP Components

Every NCP gateway may be Nation-Specific. The term "Nation-Specific NCP Components" in this document is used to identify the components of the NCP that will not be developed in common but are still part of the NCP.

As the NCP Connector connects the national infrastructure to the common components of the NCP it has to implement a defined and stable interface. While the interface must be common the implementation can vary according to the needs of the national systems.

If the national infrastructure is based on a messaging system, which is the case in at least some countries, the National Connector will contain components that are similar to the common ones and the following suggestions can be made. It is likely there will be the need of components that encapsulate functionality as described as follows:

- It is suggested to have countries re-using commonly developed components responsible for the SOAP termination. This will help to ensure interoperability of the NCPs.
- Security services: Dependent on the national implementation of system security and data protection. Assuming that certificates are used to provide confidentiality, authenticity and integrity of data this component is for example responsible to validate certificates according to the national policies.
- Message Adaptation: If a tight integration with the national systems is planned it may occur that the message sequences on the national side do not directly map to the corresponding eHealth DSI sequences. For example, it may be that certain

information in the national infrastructure is transmitted in split messages while in eHealth DSI only one message is used. Another example is that information is distributed and must be consolidated. If that is the case a component is needed which manages the mapping of the message sequences.

- Localization of medical data: the NCP will have to support a nation-specific architecture that implements localization and storage of medical documents.
- Auditing: There will be in any case the need to design and build a component which provides functionality to read audit logs according to the national requirements and legislation. In addition, there may be special requirements on the national side which produce the need for an extended auditing.
- Access control: Parts of this feature may be placed in the national infrastructure but there will be the need of a dedicated component in the national connector. At least the stored consent must be read, interpreted and used for the decision making. A structured approach for the component dedicated to access decision control has proved itself as beneficial. Thereby the component is composed of the following subcomponents:
    - The AEF (Access-Control-Enforcement-Facility) controls every data access and thereby makes sure that no access is possible without a former authorization check.
    - The ADF (Access-Control-Decision-Facility) is responsible to provide the decision whether access is granted or not. It decides based on the ADI (Access Decision Information).

Dependant on the architecture and design of the national infrastructure there may be more components needed e.g. for identification of patients or HPs but the denominated are the most likely.

3.5 Central Common Services Components

There are no technical central services providing functionality to the countries. To achieve interoperability between the NCPs centrally managed processes to maintain, update and distribute configuration data is needed.

3.6 NCP-B Internet Front-end

The NCP-B Front End will provide a simple interface for retrieving Patient Summaries (PS), ePrescriptions (eP) as well as for issuing eDispensations (eD). The existence of such a front-end can significantly reduce efforts related to piloting of country B eHealth DSI use cases. For country A uses cases no front end can be used since in this case the NCP-A has to be integrated with the national infrastructure.

3.6.1 System Overview

The NCP-B Front-End MUST be built on a Web Application Server/framework, which should be easily to maintain and minimizes development cost & complexity. It SHOULD be based on the enterprise java (Java EE) referable reduced version containing the necessary part for JSP/JSF execution (recommendation), but this can be change with justification. It MUST support the web-services Patient Identification (ID), Patient Consent (PC), patient Summary (PS), ePrescription (eP) and eDispensing (eD); consuming the first four and publishing/sending/pushing the latter.
The interface with/to the NCP MUST go through the NCP Portal Adapter, that decouples the country-B web portal and the core NCP (by this NCP and portal can be deployed in different security zones and multiple, user-group specific portals can be connected to a single NCP). Communication to and from the Front-End to NCP need the Portal Adapter, because the National Connector is unknown (national specific), and therefore to have a common/unique interface between the two systems, a portal specific adapter is needed.
The connectivity between the NCP Portal Adapter and the NCP-B Front-End can be java-to-java technology, Plain Old Java Object (POJO), but it MUST be so, that the Front-End and the Portal adapter can be separated by network zone and physical execution

environment. The transportation of POJOs can easily be facilitated with the Spring Remoting framework or equivalent.
The System overview below is informative.

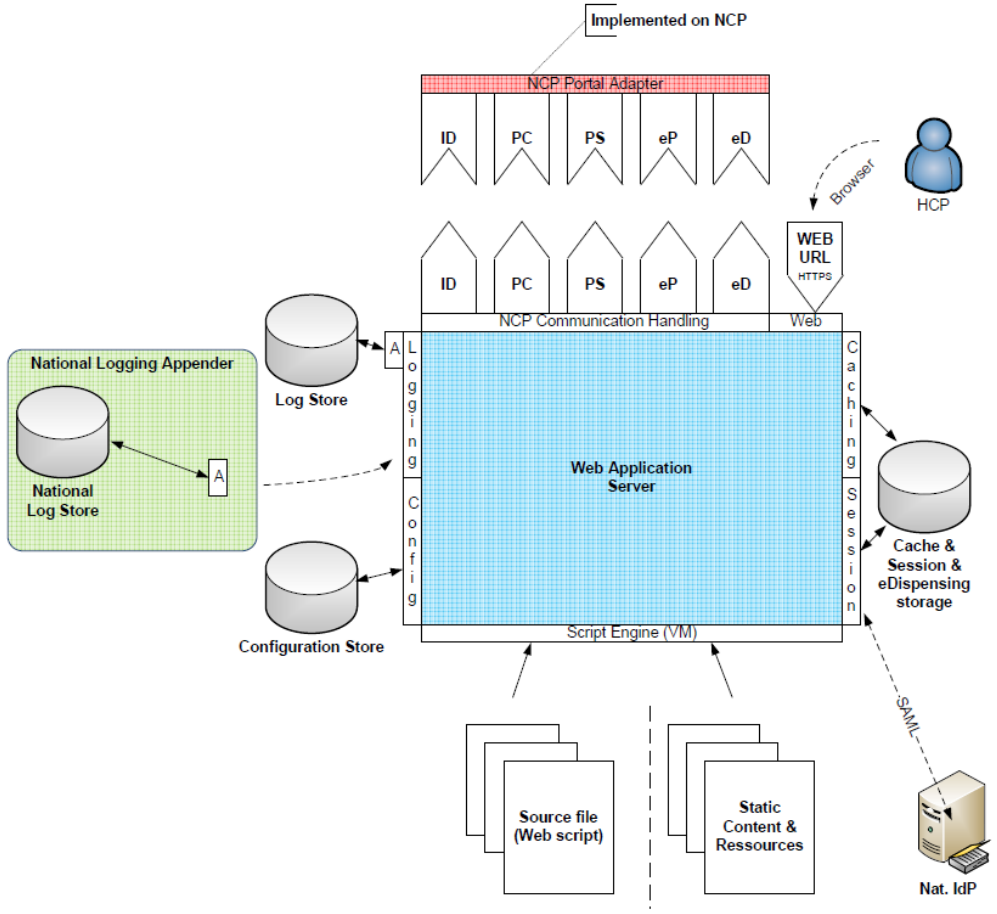


Figure 17: Front-End System Overview

The frameworks/products etc. to-be used in the Front-End development SHOULD be chosen according to the NCP Implementation Platform description.
The System overview above shows that a national logging Appender (marked with 'A') can be made, so that logging is performed, also on a national logging system. A common well-tested framework for logging with Appender (marked with 'A') setup is Log4j (recommended). Log4J supports addition/special appenders for auditing to special sources (national). Furthermore it supports auditing/logging to multiple sources, there another chosen framework, than Log4J, SHOULD support the same functionality (justification).
The standard implementation needs storage of log, configuration, cache, sessions & eDispensing rapports information (shown with dB-symbols). Furthermore source files (enterprise java ~ Java EE) must be accessed along with static content/resources at Web Application start-up.

Front-end component	**Description**
Web Application Server	Web- and/or application Server to execute the NCP Front-end B application
Script Engine	Environment for compiling, executing Web Script files and needed resources etc.
Configuration Store	Storage needed by the configuration framework of the Front-end application

Log Store	Storage needed by the logging framework of the Front-end application
National Logging Appender	A logging appender (module / component), which is not part of the Front-end delivery, but shall be a possibility for country implementers. The appender can write logging/audit entries to the national logging/audit store (standard feature in Log4J).
Cache, Session & eDispensing store	Database Storage for storing cache information, HP& Patient sessions, as well as eDispensing rapports.
National Identity Provider (IdP)	National system for identifying HP using SAML 2.0 and SAML WebSSO, not part of Front-end solution, national responsibility.
Web interface	Interface for HP access to NCP Front- end B portal, uses HTTP-over-SSL (HTTPS)
NCP communication handling	Interface to the NCP based on POJO interface by communicating Java object over-the-wire.
Portal Adapter	Adapter located in the NCP for enabling the communication between the NCP and Front-end without using the unknown National Adapters. This adapter is part of the NCP, and therefore not the Front-end.
ID	Transaction needed for enabling patient identification (NCP is provider, Front- end is consumer)
PC	Transaction needed for enabling Patient Consent documents (NCP is provider, Front-end is consumer)
PS	Transaction needed for enabling Patient Summary documents (NCP is provider, Front-end is consumer)
eP	Transaction needed for enabling ePrescription documents (NCP is provider, Front-end is consumer)
eD	Transaction needed for enabling eDispensing documents (NCP is provider, Front-end is consumer)
OrCD	Transaction needed for enabling Original clinical documents (NCP is provider, Front-end is consumer)

3.6.2 Security

The web application MUST whenever sessions are used, use Server-Side session, which are invisible to the client. Furthermore, the application SHOULD use standard web-server security measures.

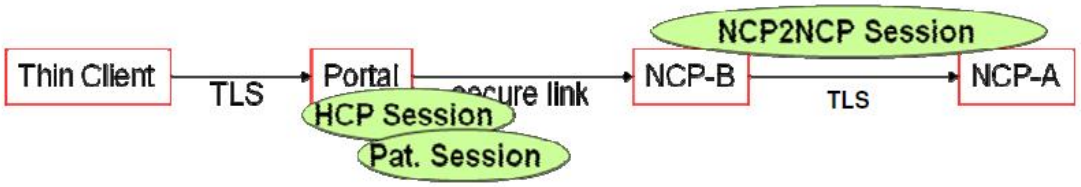


Figure 18: Security of Front-End solution (portal)

The user browser access to the portal MUST be only acceptable by HTTPS/TLS with at least 128bit encryption. This MUST be implemented with a country specific certificate which is certifiable (can be checked by national certificate authorization (CA)).
The portal MUST use a secure link to national side of NCP, which MUST be terminated in the NCP Portal Adapter (see Figure 1 (NCP-in-a-transparent-box)).

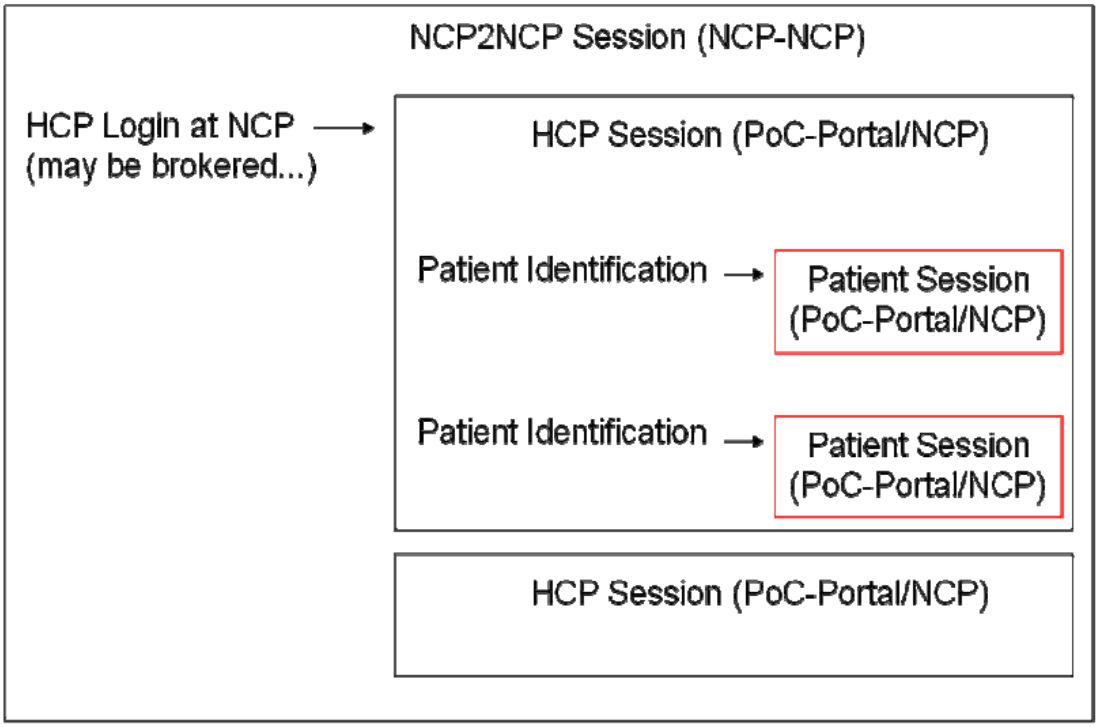


Figure 19: Session management in country B

The session needed for NCP & Front-end implementation according to the eHealth DSI standard are a NCP2NCP session (Mutual Trust), a HP Session for each HP Authentication (HP Confirmation Assertion), which can be brokered via the National IdP & a patient Session for each identified patient (TRC Assertion), which shall not be overlapping in each HP session, to provide patient security by isolating patient data.

3.6.3 HP Authentication

HPs acting as a PoC in a country B shall have the option to use the web portal to access eHealth DSI services. Strictly speaking, this is part of the national infrastructure, but as that part of the NCP is useful to a number of countries, it will be provided as part of the NCP-In-A-Transparent-Box. With respect to Identity Management, the interface will be based on the SAML industry standard, more specifically on the Web-SSO profile. The rationale for that is:

- SAML is the most commonly used standard for the Web-SSO use case, and there are several quite well interoperable implementations.

- If there is no existing Identity Provider (IdP) for HPs in a country, there are several solid open source implementations available, that can be used for operation and NCP development.
- SAML-based IdP's might be used for web service authentication as well with co-existence of browser and WS-base access in a single solution.
- The complexity of various authentication mechanisms is delegated to an IdP external to the NCP

The NCP-B Front-End MUST use the SAML 2.0 for standard authentication and assertion.

3.6.3.1 Use Case

The Web-SSO use case delegates the management of use authentication, attributes and authorization to one or several Identity Providers. The following constraints shall limit the complexity of implementation and increase interoperability. There might be separate authorities for HP authentication and HP attributes (like who is a GP). The use case assumes, that HP attributes are available to the IdP in a secure way (web service, LDAP over SSL, etc.) and issued by the IdP as SAML attribute assertion.

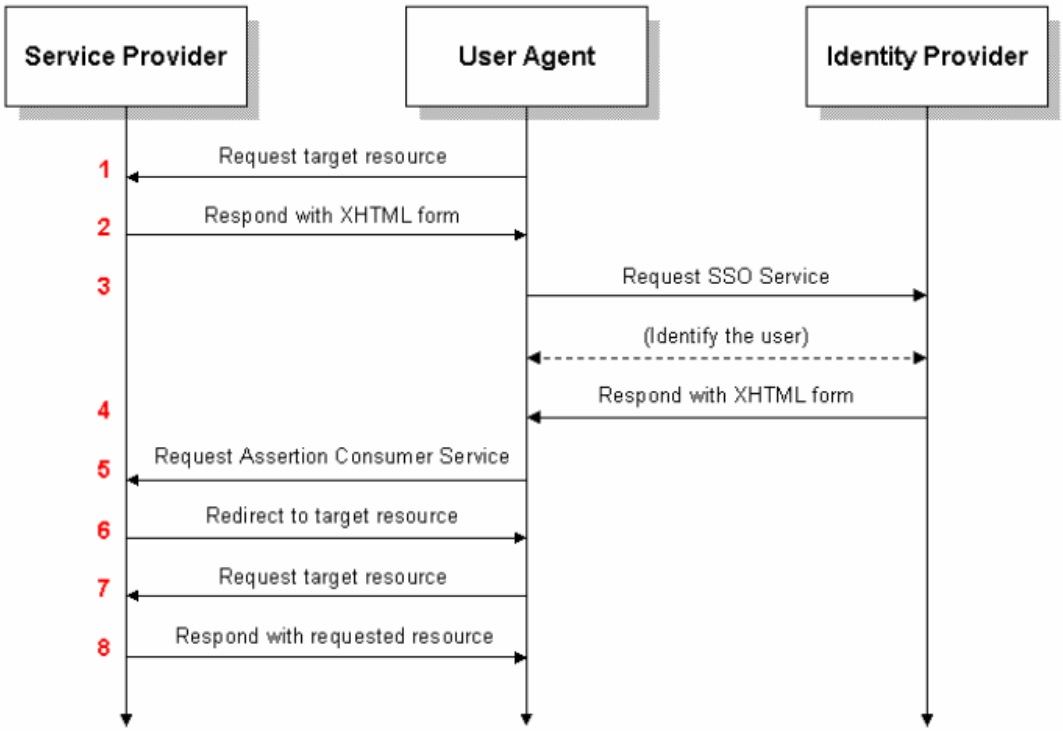


Figure 20: SAML Web SSO Sequence Diagram

The SAML Assertion received from the Identity Provider (IdP) MUST provide the eHealth DSI attributes needed for issuing the HP Identification SAML assertion.
The flow control is as follows:

1. The User Agent (i. e. Web Browser) calls the Service Provider (i.e. NCP Frontend).
2. The Service Provider responds with a XHTML form containing the localization information of the trusted Identity Provider.
3. The User Agent performs the authentication of the logged in user.
4. The Identity Provider responds with a XHTML form that contains the SAML Assertion.
5. The User Agent calls the Service Provider again, but with the issued assertion.
6. The Service Provider accepts the SAML Assertion and responds with a redirect to the target resource.
7. The User Agent requests the resource that is provided by the Service Provider.

8. The Service Provider responds with the requested resource.

Mapping of terms in figure 14 to eHealth DSI Front-end terms.

SAML WebSSO Term	NCP Front-end equivalent
Service Provider	Front-end – Web Interface
User Agent	HP Browser for Front-end access
Identity Provider	National Identity Provider (IdP)

3.6.3.2 NCP Web-Portal SAML profile

- Only SAML 2.0 needs to be supported, SAML 1.1. is not considered
- Authentication is always initiated from the SP; IdP-discovery can be any type
- Authentication Request with both HTTP Redirect and HTTP-Artifact bindings
- All IdP-SP communication uses HTTP
- Request may be unsigned

Authentication Response:

- Binding HTTP POST
- Assertion must be signed
- Attributes should be contained in the authentication response in one assertion in a single AttributeStatement

Attribute Query

- SP Authentication shall support HTTPS (signed requests are optional)

HP Attributes

- All attributes are available to the IdP as attribute authority, and can be queried by the SP using an SAML query

Single Logout (SLO)

- SP-initiated SLO must be and IdP-initiated SLO should be supported
- LogoutRequest and LogoutResponse MUST be signed
- If Single Logout is unsuccessful, user MUST be informed

3.6.3.3 SAML Authentication Context Classes

Initially, X.509 client certificate based authentication is required. IdP and SP MUST ensure that the client is using the same certificate for authentication and resource access. Otherwise, there is no restriction on client authentication mechanisms.

3.6.3.4 Metadata Distribution

IdP meta data must be signed for operational use.

3.6.4 Site Collection

The NCP-B Front-End site/portal pages SHOULD have the structure shown below, that is the navigation tree is informative, but functionality within SHOULD be present.

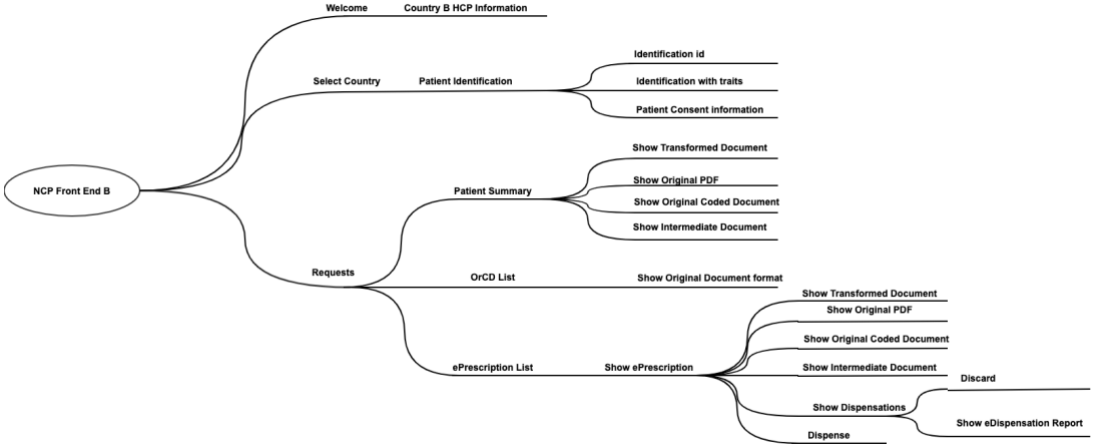


Figure 21: Site diagram (Site Collection)

The pages in the site-collection are not listed in detail, but should be self-explanatory with little knowledge with eHealth DSI; however, the clinical information pages are explained in the following table.

Page name/type	Description
Show Transformed Document	Page showing the structured transformed data coming from the Pivot CDA document
Show Original PDF	Page/link to CDA embedded PDF/A document (launch of browser registered PDF viewer)
Show Original Coded Document	Page showing the structured original data coming from the Pivot CDA document. The data is unhampered by the transformation processes in the two communicating NCPs.
Show Intermediate Document	Page showing the structured eHealth DSI intermediate data coming from the Pivot CDA document. The data is in the eHealth DSI coding & eHealth DSI language (English)

3.6.5 Application Life-Cycle

The NCP-B Front-End application MUST have an application life-cycle, which is checked by the application, so that a controlled start and stop is performed. Each state can only be entered if the previous action was successful. The life-cycle below is informative.

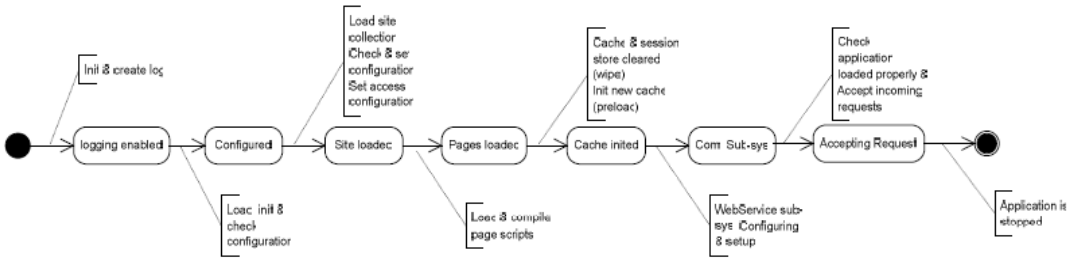


Figure 22: Application Lifecycle (Informative)

An important step in above is the clearing/wiping of cache & session stores; to avoid information creep between sessions etc. These are for security normative and MUST be performed before accepting of client requests.

3.6.6 Page Life-Cycle

The NCP-B Front-End application MUST have an application/web page life-cycle, which is checked by the application, so that page loading and request/response is controlled. Each state can only be entered if the previous action succeeded. The life-cycle below is normative, showing that Front-End requests to NCP are performed in parallel to page showing a 'Please Wait'-sign. This sort of parallel execution CAN e.g. be performed in AJAX (Client-Side) or Server-Side. This life-cycle start MUST only be accessible from Application life-cycle state: "Accepting Requests".

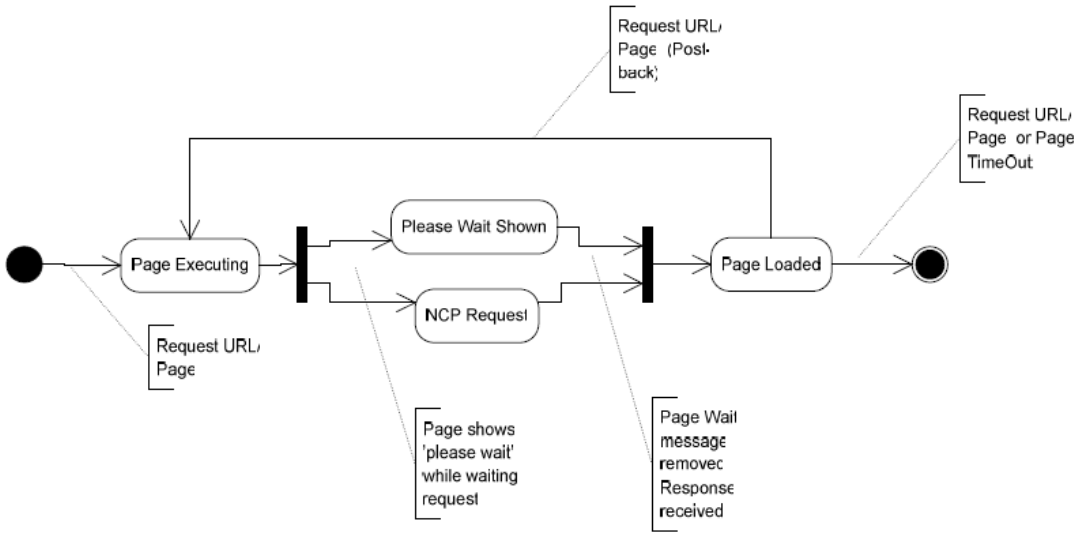


Figure 23: Page Lifecycle (NCP request handling)

3.6.7 HP Session Life-Cycle

The NCP-B Front-End application MUST have an HP Session life-cycle, which is checked by the application, so that HP authentication, access-control & timeouts are controlled. Each state can only be entered if the previous succeeded. The life-cycle below is normative, showing that HP authentication MUST be done before actual patient identification can be performed. This life-cycle start MUST only be accessible from Application life-cycle state: "Accepting Requests".

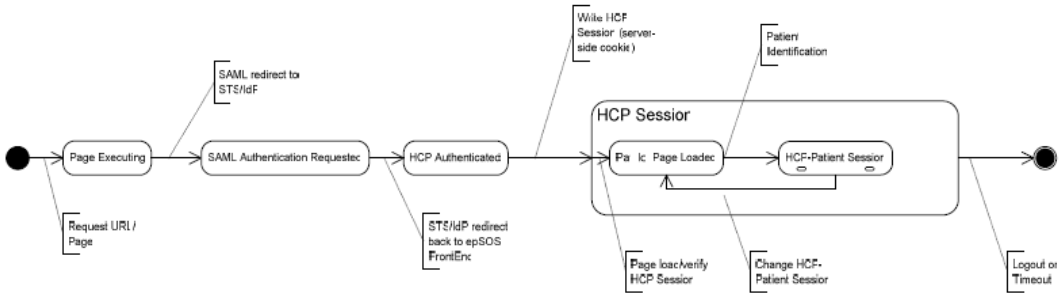


Figure 20: HP Session Lifecycle

The HP Session is stored on the Front-end Web Server (Server-side), as an entry/object in a session store for controlling access and state of HP session.

The authentication MUST be via direct- or brokered trust SAML assertions. This MUST be done using the SAML 2.0 Web Single-Sign-On (SSO) profile. Below this is shown on the right. On the left a deployment is shown, where HPs directly authenticate with the NCP.

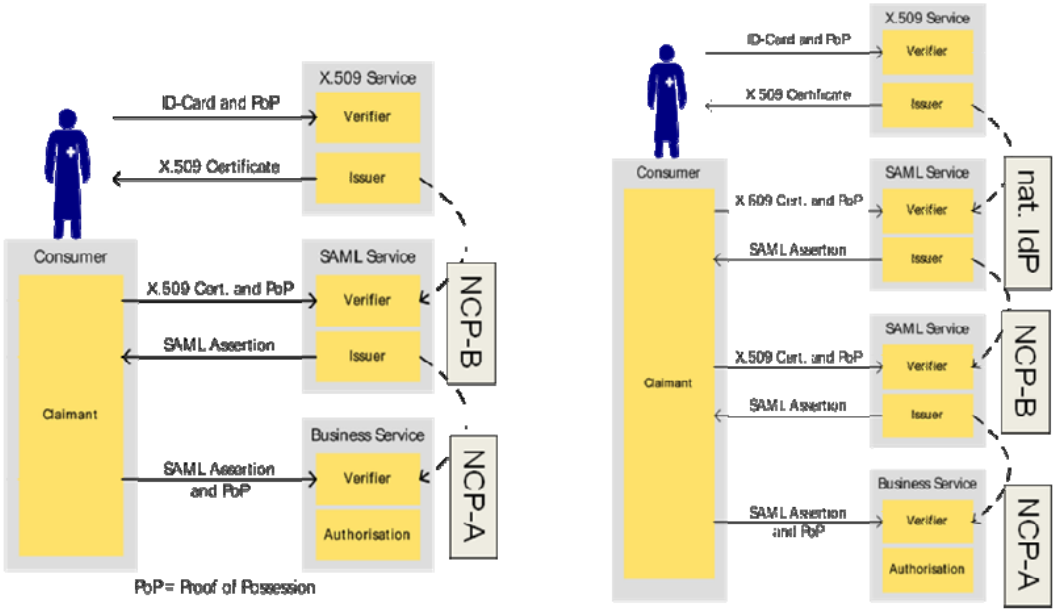


Figure 21: Example from NCP assertion trust

The IDP can only issue identity assertions; the TRC is COMPILED (content gathering) by the CLIENT and IS SIGNED by TRC-STS.

3.6.8 HP Patient (TRC) Session Life-Cycle

The NCP-B Front-End application MUST have an HP Patient/Treatment Relationship Confirmation (TRC) Session life-cycle, which is checked by the application, so that an authenticated HP with an identified patient can access patient data in a controlled manner. Each state can only be entered if the previous action was successful. The life-cycle below is normative, showing that HP MUST perform patient identification before patient data can be retrieved. This life-cycle start MUST only be accessible from HP Session life-cycle state: "HP-Patient Session".

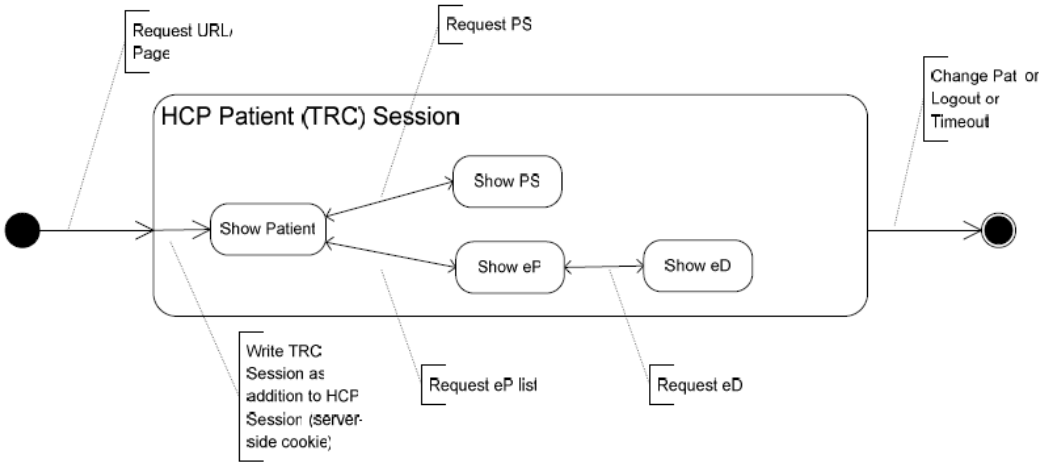


Figure 22: HP Patient (TRC) Session Lifecycle

The handling of multiple session and cascaded session in the Front-end application must be manually implemented if Java environment does not support this, depending on versions hereof.

4 Example Deployment Composition

Since all major actors or components inside the NCP-In-A-Transparent-Box are introduced in previous chapters, now a closer look how they might be deployed is given.

Introductory, the figure below depicts the placement of both NCPs. A Thin Client (Browser-based eHealth DSI Portal or Front-end) enforces an authentication via a trusted Authentication Service, i.e. Identity Provider. The protocol used (either SAML Protocol or WS-Trust Protocol) is out of scope for the NCP implementation. Only the outcome of the Identity Provider – the (signed) SAML assertion (Identity Assertion) - issued by a trusted party located in the national infrastructure - is relevant. The eHealth DSI Portal invokes the NCP-B providing all specified eHealth DSI services. The SOAP / FHIR requests are fully compliant to the eHealth DSI normative request message specification. The NCP-A responds with a message according to the specified eHealth DSI message format.
Figure 23 shows the proxy semantic of a NCP-B. That is, a generic Java-based invocation message is transformed into the eHealth DSI format and forwarded to another NCP-A, and finally mapped on national specific service requests. For simplicity, only relevant components are shown, those which have an impact on the communication with involved services or other components. The flow of control with regard to the common components of the NCP is given in the next section.

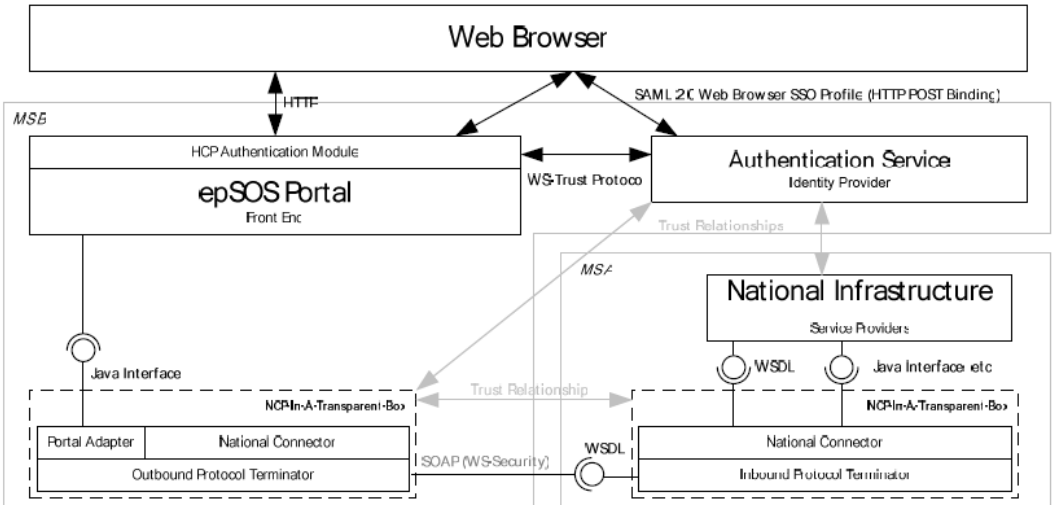


Figure 23: Logical View for both NCPs[2] (Note, the protocol terminators are removed, this picture needs updating)

4.1 Navigation Paths

Since the NCP takes different roles in the Circle of Trust, the navigation paths for the message flow are exemplary shown below. Therefore a NCP provides three interfaces which are also described. These ones are the (1) NCP-B Interface, the (2) NCP-A Interface, and the Central Common Service Interface.

4.1.1 NCP-B Role

The eHealth DSI portal invokes a method provided by the portal adapter interface of the National Connector. The portal adapter forwards the request to the Workflow Manager, that makes use of various common components to process the request:

- the provided assertions need to be validated, e.g. it must be checked whether the eHealth DSI profile is used
- routing information needs to be obtained from the eHealth DSI NCP configuration
- audit trail entries need to be written
- for eDispensations a mapping of the national format and taxonomies to the eHealth DSI format must be done. For ePrescriptions and Patient Summaries a mapping from the common eHealth DSI format to the national format and vocabularies must be done

[2] This figure shows a sample deployment. E.g. instead of WS Trust the SAML protocol could be used as well for obtaining the HP identity assertion.

Finally, a SOAP request message is created and sent to the cross-border NCP-A. The response is mapped on the interface of the National Connector so that the eHealth DSI Portal can render it accordingly.

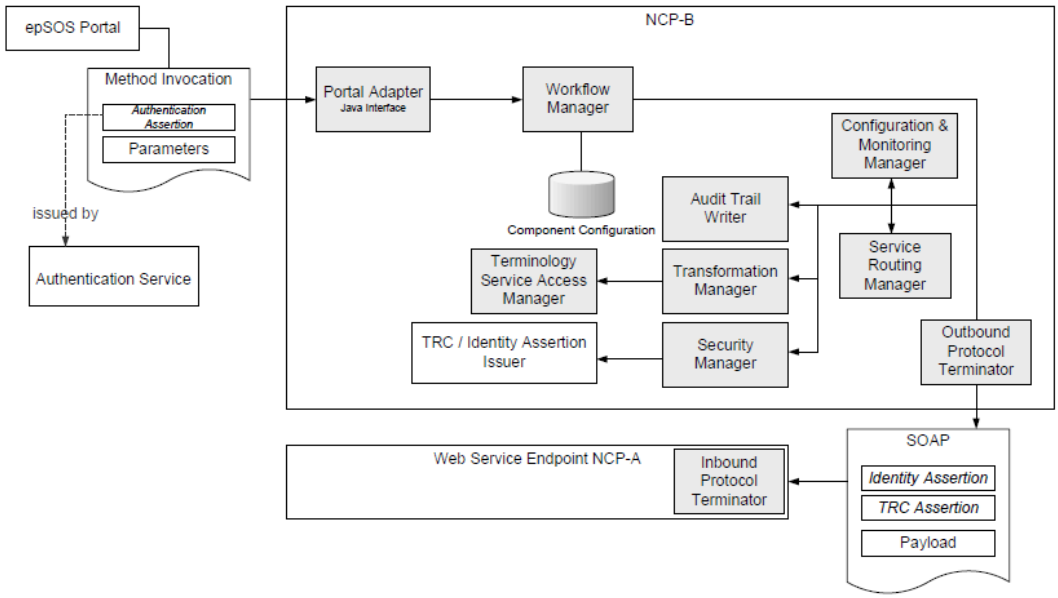


Figure 24: Overview for NCP-In-A-Transparent-Box in NCP-B Role

The eHealth DSI Front-end is logically separated from the NCP-In-A-Transparent-Box. However, it is tightly bound to it, because the Front-End provides the user interface for the eHealth DSI services which the NCP delivers or supplies. What is solely taken into account when regarding the NCP-B role is the communication between the Front-end and the National Connector that the NCP-B provides. Until now, the interface to be specified is an ordinary Java one, which accepts POJOs (Plain Old Java Object) as parameter values. Further information on the technology that implements the serialization and de-serialization of them is given in section 3.6.1.

4.1.2 NCP-A Role

The incoming SOAP messages are processed by off the shelf Web service frameworks (cp. section 5.21). A registered Web service endpoint listener is called by the framework. Previously, the SOAP message was validated against the policy according to the WSDL document. For instance, this includes the token mapping (i.e. are SAML assertion(s) and signatures present), or the validation of timestamps stated in the message header. SOAP message handlers provide the possibility to access the entire SOAP envelope and might pre-process relevant data. The Endpoint Listener issues synchronous (blocking) requests to the Workflow Manager component. Then, this component uses registered components in order to process the request. In the end, the services located in the national infrastructure are invoked by means of the National Connector in order to create the SOAP response for NCP-B. The following figure sketches the role of a NCP when it is called by another NCP-B.

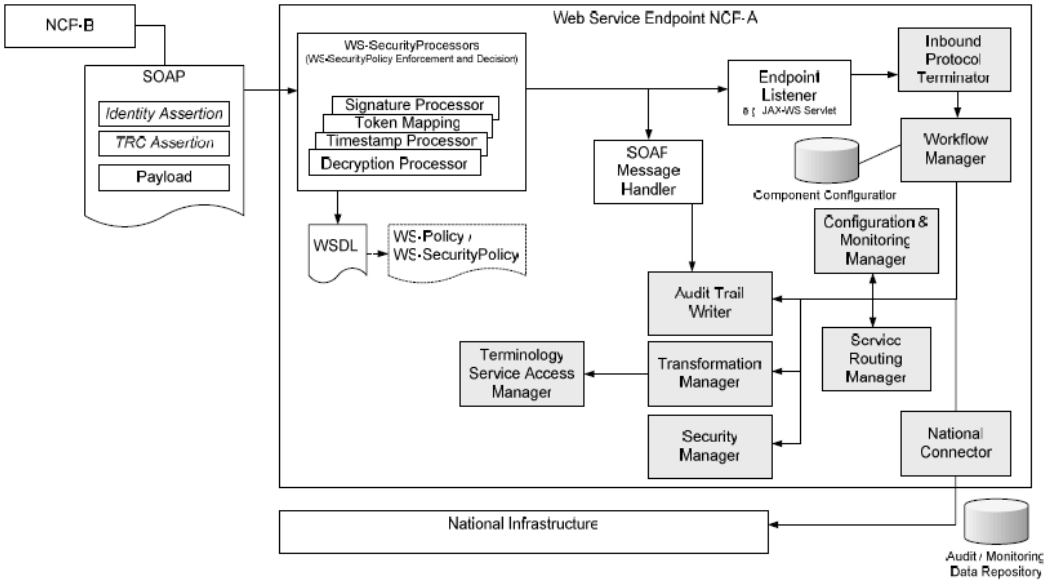


Figure 25: Overview for NCP-In-A-Transparent-Box in NCP-A Role

4.1.3 Central Common Service Role

There exist several external services that are consumed by a NCP implementation in order to process a request. Regardless of the origin of the configuration (e.g. document or dedicated services), each NCP has to be initialized and put into operation it in a normalized manner. This necessitates that a common interface is available for relying components inside the NCP. Section 3.2.10 introduced the Configuration and Monitoring Manager which encapsulates the source of the configuration. The access to the configuration data needs to be defined subsequently.

4.2 Deployment

Comments to the deployment diagram:

- Segregation of duties requires that the audit repository and NCP service MUST run in separate administrative domains. This does not require separate physical servers, but safeguards, that the NCP administrative staff does not have access to the audit repository.
- Transport layer security configuration is dependent on whether a web service proxy must be installed. (Opinion: Using a web service proxy makes certificate management more cumbersome, for a very small security advantage.)

There are no specific hooks for monitoring services at the application level; (Beside an embedded anomaly detection system signaling if exchanges are exceeding a parametrized threshold). The following figure sketches the exemplary deployment of the common components in a NCP and its related services used.

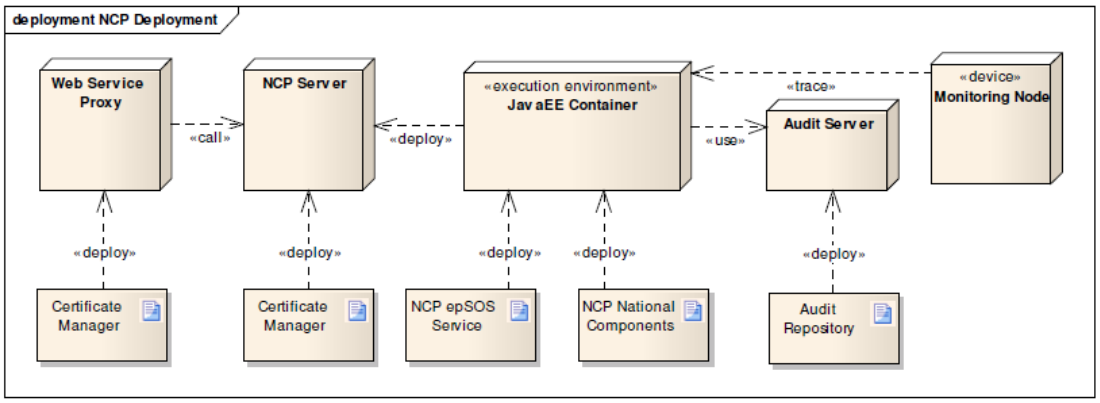


Figure 26: Exemplary Deployment of the NCP Components

4.3 Implementation Platform

eHealth DSI Common software will adopt Open Source frameworks and applications as much as possible.

Reference implementations will be made available on a single platform. To open vendors a wide range of options, the only two requirements are:

- RedHat Enterprise Linux
- Java Enterprise Edition

Vendors will have to specify the dependency of their solution on languages, frameworks, APIs and applications; the detailed specification will contain a sample list. All choices of frameworks, libraries application environment must be compatible as specified from source and by vendor test.

The goal of the following table is to provide a first list of software languages, frameworks and applications for eHealth DSI software commonly developed components and for eHealth DSI commonly developed Front-end.

Element	Required	Options / choices / Recommendation
Languages	Java 17 (TBD 21)	Oracle JDK or OpenJDK
Operating System		Ubuntu Linux (64bit) Choice between version 4 or 5
Application Server		Recommended choice between • Tomcat v.8.x • Oracle WebLogic 12c
GUI		Recommended build upon JSP/JSF
Security SAML Framework		Recommended • OpenSAML library
Security WS-* Framework		Choice between • SUN METRO • Apache CXF
Audit Framework		Recommended using log4j for audit logging

4.4 Security Setup for NCP

4.4.1 Security Zones

This section is informative to the country, however likewise security mechanisms MUST be adopted by the country in accordance with [Security Services Specification].
Figure below from [Security Services Specification] shows the security zones for eHealth DSI setup of the NCP.

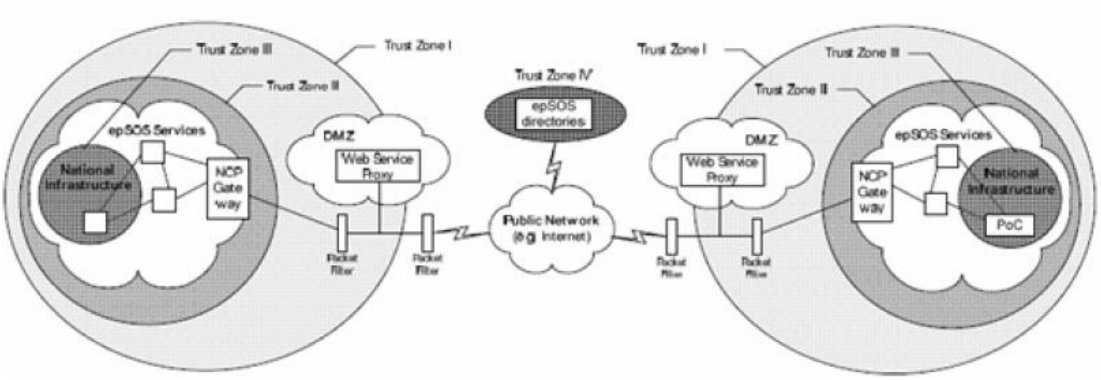
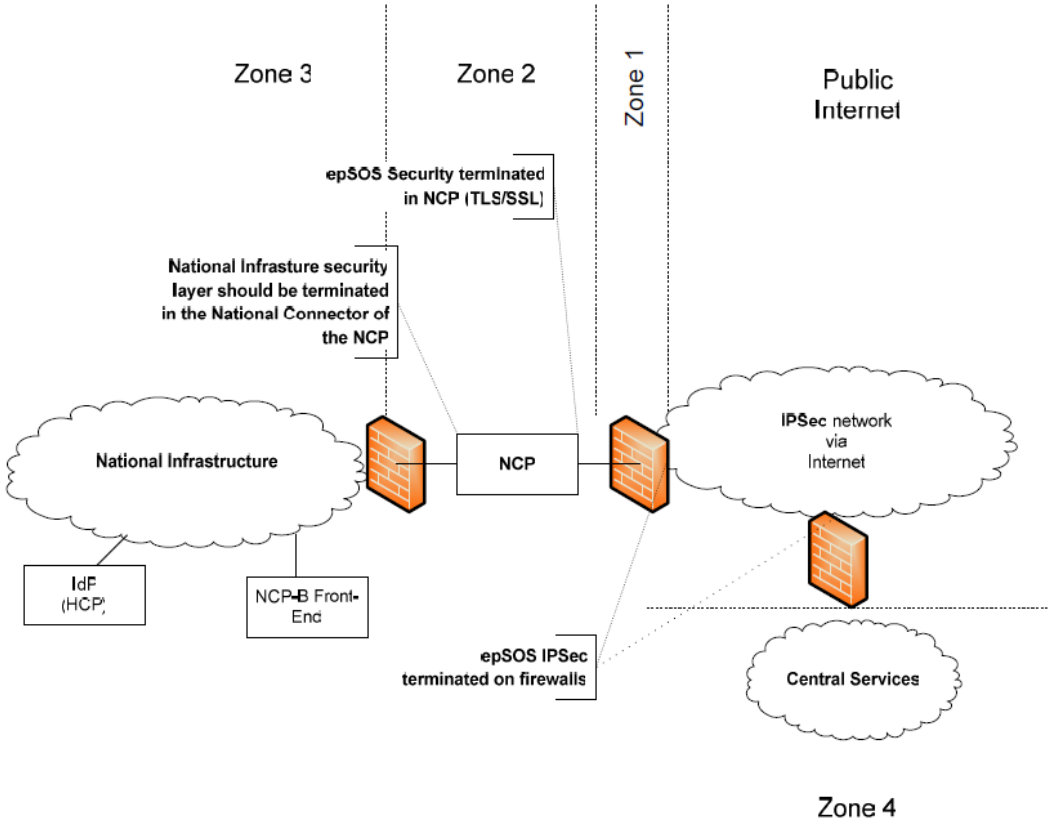


Figure 27: General eHealth DSI Trust Zones

Adaptation of the security zones to the "NCP in a transparent box" approach, i.e. the NCP is a gateway (box), which is seen as both secure (national) and un-secure (entrance of internet traffic). To strengthen the protection both for the national infrastructure and the traffic entering the eHealth DSI p2p network, the NCP is placed in its own security zone, guarded by two firewalls.



System / Zone	Description	Security Mechanisms
eHealth DSI Central Services	Central Services placed in the eHealth DSI p2p network (Internet), should be guarded by the same security as the NCP zone, since the information is used by NCP. The Central Cervices might be implemented as virtual following the p2p paradigm and there would physical be placed in the NCP zone. Virtual central services are seemingly at a central place, but actually physically at the NCPs (decentralized). Only the reference mechanism is central (referenced by DNS).	Same as NCP gateway and zone.
NCP gateway	NCP gateway in its own zone. The NCP MUST terminate the layer 4+ (SSL/TLS) on the NCP gateway platform inside the NCP gateway process / execution environment. This should be the case for national infrastructure security also, meaning that no unencrypted patient data SHOULD be disclosed in the NCP ZONE outside the NCP gateway (impact on gateway application design), see figure below. NCP contains PDP, PEP & STS. NCP zone should be on the inside of public DMZ zone in country setup	Hardened Platform
Firewall towards internet	Firewall that provides protection for the NCP and indirectly the national infrastructure. It should be a hardened platform. It can be implemented as more than one system.	Hardened platform Address, Port, Packet filtering Stateful Web Service proxy Active security mechanisms & intrusion detection
Firewall towards nat. inf.	Firewall that provides protection for the national infrastructure and as a filter protection for the eHealth DSI network	Hardened platform Address,Port,Packet filtering

Firewall (centr.serv.)	Firewall that provides protection for the eHealth DSI central services, which should provide the same security level as the "Firewall towards Internet" The firewall is redundant if the central services are virtual, i.e. follow the p2p pattern, and the central services are placed in the NCP-zone or better.	Sameas"Firewall towards Internet"

4.4.2 NCP gateway Application

The NCP MUST terminate the layer 4+ (SSL/TLS) on the NCP gateway platform inside the NCP gateway process / execution environment. This should be the case for national infrastructure security also, meaning that no unencrypted patient data SHOULD be disclosed in the NCP ZONE outside the NCP gateway (impact on gateway application design), see figure below.

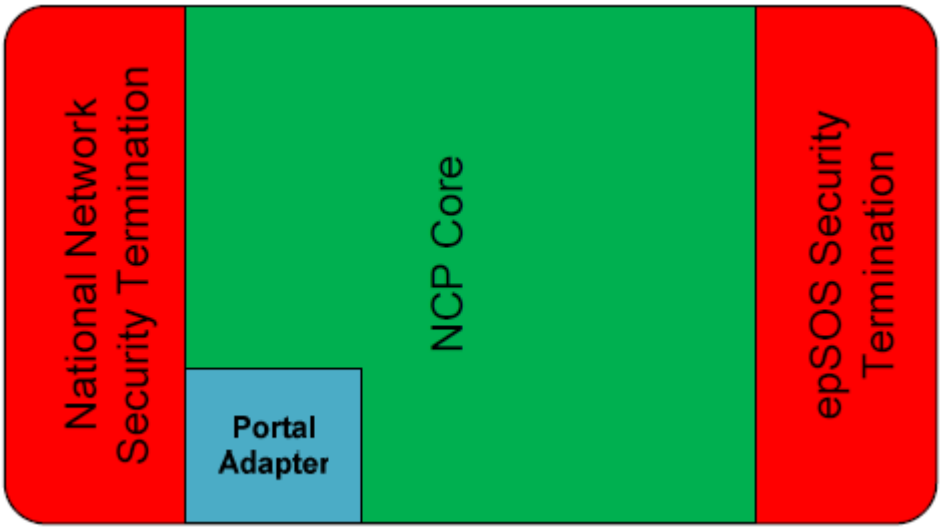


Figure 28: NCP Application Security

The figure shows the NCP Gateway application process, which needs to include the national & eHealth DSI security termination to secure that no disclosure of data is done out-side the applications security domain (secure memory). It is recommended, that the NCP gateway is started in a secure way, EX: By start with user account, that changes password on start-up.

5 Components Design

This section contains the individual description of the components and a further decomposition of the solution for the NCP.

5.1 Services

5.1.1 FHIR implementation

The client and server provide a REST endpoint collection to support FHIR based resource calls. The HapiHL7WebServer is used as the FHIR implementation. The wrapped security assertions are wrapped inside a JWT token.

5.1.2 eHealth DSI Patient Identification Service

In order to discover a patient's data, the patient must be identified, and a unique patient identifier must be shared between the communicating gateways. This shared identifier enables the medical data consuming services to properly reference that patients' data which is provided by the medical data providing services at the patient's country of affiliation.

The eHealth DSI Patient Identification Service defines means for the agreement on this shared identifier and for increasing the degree of accuracy of the patient identification means that are used within the country of care. This shared identifier MUST be used as a patient identifier as required by the eHealth DSI medical data exchange services (e. g. patient service and order service).

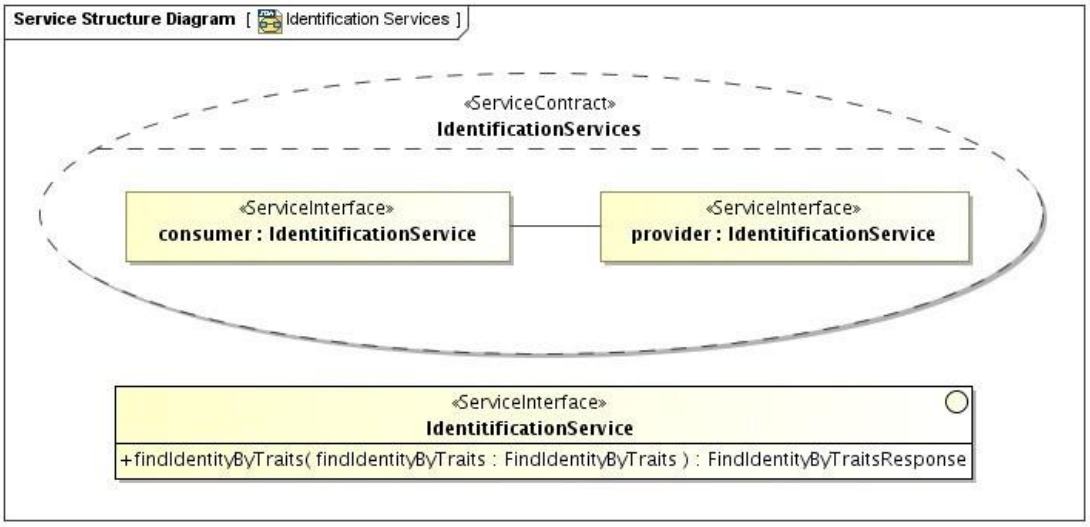


Figure 30: eHealth DSI Patient Identification Service Interface

Operation	findIdentityByTraits()	
Description	Obtain a shared patient identifier	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	FindIdentityByTraitsRequest	
	Body	(1) List of patient identity traits as provided by the patient to the HP. (2) optional: minimum confidence level that has to be met by the entities that match the provided traits.
	Security Token[3]	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion
Output Message in successful Case	FindIdentityByTraitsResponse	
	Body	(1) Unique identifier of the patient that has to be used for all sub-sequent calls for this patient's medical data. (2) optional: further patient identity traits that allow the HP to verify the result of this operation.

[3] PT = Protection Token, ST = Supporting Token (according to [WS SecurityPolicy] definition of security token types)

		If no unique match is found, the service provider MAY respond with a list of candidates. For each candidate body elements (1) and (2) MUST be provided.
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer is able to locate the service provider 2. A secure channel is established between service consumer and service provider nodes 3. The service provider is able to verify the authenticity of the service consumer 4. An HP identity assertion has been issued for the requesting HP 5. The service provider is able to verify the validity of the HP identity assertion 6. the service consumer is able to verify the authenticity of the service provider 7. The patient has given a consent that authorises NCP-A to disclose his identity	
Main success scenario	Actions of the *eHealth DSI Identification Service* provider: 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion 3. Verify that the requesting HP is authorised to query for patient IDs 4. Extract the patient identity traits from the message body 5. Search for patients that match the provided ID attributes 6. depending on the number of matches: - If multiple patients match: request for more identity traits or provide a list of candidates (depending on national security policy). If a list of matching candidates is provided it MUST only include patients who gave consent to ehealth DSI. - If single patient matches and this patient has given consent to eHealth DSI: select ID to be used for subsequent requests - Any other case: throw respective fault 7. Apply eHealth DSI protection means to the response message and send it to the requestor	
Fault Conditions	Preconditions for a success scenario are not met	
	Requesting HP has insufficient rights to query for a patient's identity	
	No matching patient is discovered that gave consent to eHealth DSI	
	ID traits are insufficient for country A to find a matching patient (e.g. provided search criteria are not supported)	
	The confidence level of the matches is too low with respect to the level required by the requestor	
	Patient identification is only performed in conjunction with patient authentication (e.g. by providing a secret or a reference to a valid eIDAS authentication)	
	Confirming the query would lead to a privacy violation acc. to country A legislation.	

5.1.3 eHealth DSI Patient Service

The eHealth DSI Patient Service provides operation for retrieving an identified patient's patient summary.

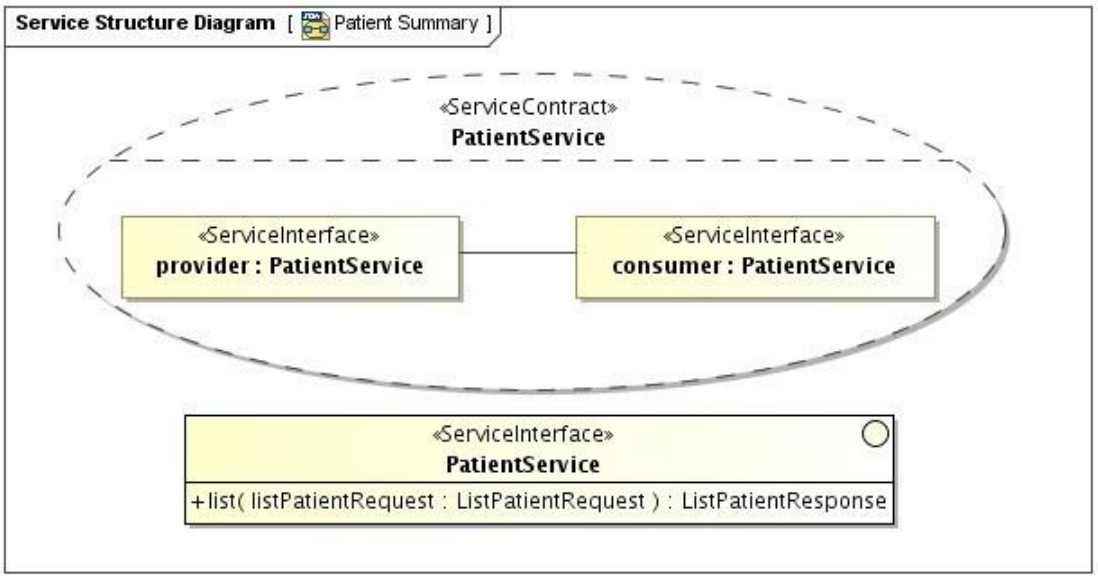


Figure 31: eHealth DSI Patient Service Interface

Operation	list()	
Description	Obtain the patient summary of the identified patient	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	ListPatientRequest	
	Body	(1) Identifier of the patient whose patient summary is requested (2) Optional: eHealth DSI CDA template qualifier (pivot and/or source coded document[4]). If no template qualifier is given the service provider MUST provide all available encodings.
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion [ST] eHealth DSI Treatment Relationship Confirmation Assertion [O]
Output Message in successful Case	ListPatientResponse	
	Body	(1) eHealth DSI-encoded patient summary (CDA) or/and (2) source coded patient summary of the identified patient
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer can locate the service provider 2. A secure channel is established between service consumer and service provider nodes 3. The service provider can verify the authenticity of the service consumer 4. service consumer and service provider share a common identifier for the patient 5. The patient has given consent to the use of eHealth DSI 6. A valid patient summary for the identified patient is accessible for NCP-A 7. The requesting HP has been authenticated in the country of care and the service provider is able to verify the attesting HP identity assertion 8. A treatment relationship exists between the patient and the requesting HP and the attesting assertion can be verified by the service provider 9. the HP is authorised to access the requested data 10. the service consumer is able to verify the authenticity of the service provider	
Main success scenario	Actions of the eHealth DSI Patient Service provider: 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion and TRC assertion 3. Verify that the patient has given valid consent to eHealth DSI and that the consent applies to the	

[4] In this document the term „source coded document“ is used for an encoding of a medical document that did not undergo any semantic translation. Following [Semantic Services Specification] it is assumed that PDF/A embedded within CDA is used as the format of choice for source coded documents.

	current usage scenario 4. Retrieve patient's patient summary source document 5. Enforce national security policy and (if available) patient privacy policy 6. Verify authenticity and integrity of the patient summary 7. Transform patient summary into eHealth DSI pivot format (if requested and needed) and write a respective audit trail entry (see section 4.5.7.3) 8. Render PDF from source document (if requested and needed) 9. Apply eHealth DSi protection means to the response message and send it to the requestor
Fault Conditions	Preconditions for a success scenario are not met
	Requestor has insufficient rights to access the patient's medical summary
	No patient summary is available for the identified patient
	No consent for patient summary sharing is registered for the identified patient
	The patient summary cannot be provided in the requested encoding
	Temporary failure (e. g. verification of preconditions cannot be performed due to a system failure)
Warning Conditions	Country A allows for data hiding; a respective disclaimer SHOULD be shown to the HP
	The HP MUST additionally consider the source coded document because this MAY contain additional information
	The computation of the eHealth DSI encoded patient summary was not approved by an HP; a respective disclaimer MUST be shown to the requesting HP
	The original data (provided as source coded document) was totally or in parts assembled automatically and has not been approved by a HP; a respective disclaimer MUST be shown to the requesting HP

5.1.4 eHealth DSI Order Service

The eHealth DSI Order Service provides operation for retrieving an identified patient's available ePrescriptions.

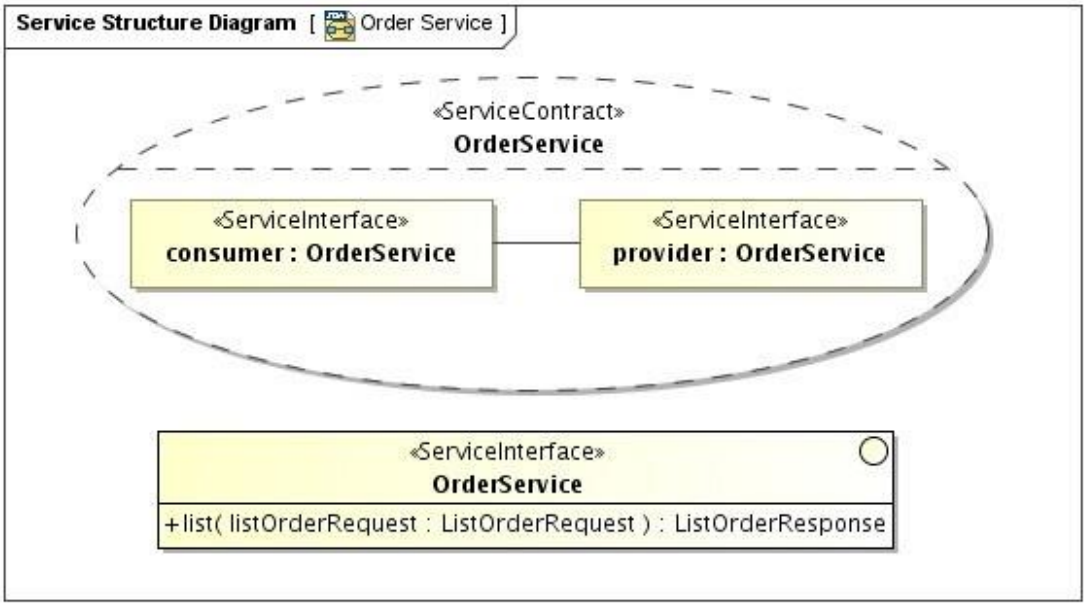


Figure 32: eHealth DSI Order Service Interface

Operation	list()	
Description	Obtain the eHealth DSI-encoded, available ePrescriptions of the identified patient	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	ListOrderRequest	
	Body	(1) Identifier of the patient whose available ePrescriptions are requested (2) Optional: eHealth DSI CDA template qualifier (pivot and/or source coded documents). If no template qualifier is given the service provider MUST respond with all available encodings of the re- quested documents.

	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion [ST] ehealth DSI Treatment Relationship Confirmation Assertion [O]
Output Message in successful Case	ListOrderResponse	
	Body	(1) List of (1a) eHealth DSI-encoded ePrescriptions and/or (1b) source coded ePrescriptions (acc. to requested format) Of the identified patient
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer is able to locate the service provider 2. A secure channel is established between service consumer and service provider nodes 3. The service provider is able to verify the authenticity of the service consumer 4. service consumer and service provider share a common identifier for the patient 5. The patient has given consent to the use of eHealth DSI 6. All available ePrescriptions for the identified patient are accessible for NCP-A 7. The requesting HP has been authenticated in the country of care and the service provider is able to verify the attesting HP identity assertion 8. A treatment relationship exists between the patient and the requesting HP and the attesting assertion can be verified by the service provider 9. the HP is authorised to access the requested data 10. the service consumer can verify the authenticity of the service provider	
Main success scenario	Actions of the eHealth DSI Order Service provider: 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion and TRC assertion 3. Verify that the patient has given valid consent to eHealth DSI and that the consent applies to the current usage scenario 4. Retrieve patient's available prescriptions source documents 5. Enforce national security policy and (if available) patient privacy policy 6. Verify authenticity and integrity of available prescriptions 7. Transform ePrescriptions into eHealth DSI pivot format (if requested and needed) and write a respective audit trail entry (see section 4.5.7.3) 8. Render PDF from source document (if requested and needed) 9. Apply eHealth DSI protection means to the response message and send it to the requestor	
Fault Conditions	Preconditions for a success scenario are not given	
	Requestor has insufficient rights to access the patient's ePrescriptions	
	No dispensable ePrescriptions are available for the identified patient	
	No consent for ePrescription sharing is registered for the identified patient	
	A (referenced) ePrescription cannot be provided in the requested encoding	
	Temporary failure (e. g. authenticity verification cannot be performed due to a PKI failure)	
Warning Conditions	Country A allows for data hiding; a respective disclaimer SHOULD be shown to the HP	
	Country A legislation assigns the task of checking for contraindications and drug interaction to the dispenser. A warning MUST be shown to the HP in country B that the checks for contraindication and drug interaction MAY have not been performed on the prescribed medication in country A.	
	Mandatory fields have been nullified for some of the provided ePrescriptions (minimum dataset is not fully provided); the HP MUST additionally consider the source coded document	
	More ePrescriptions MAY be available but are not accessible	
	The computation of the CDA encoded ePrescription documents was not approved by an HP; a respective disclaimer MUST be shown to the HP	
	A (referenced) ePrescription is time valid for dispensation but not for reimbursement. A message SHOULD be shown to the HP that he SHOULD inform the patient that his health insurance will not reimburse the dispensed medicine.	

5.1.5 eHealth DSI Dispensation Service

The eHealth DSI Dispensation Service provides operations for notifiying the patient's country of affiliation on the dispensation of an ePrescription.

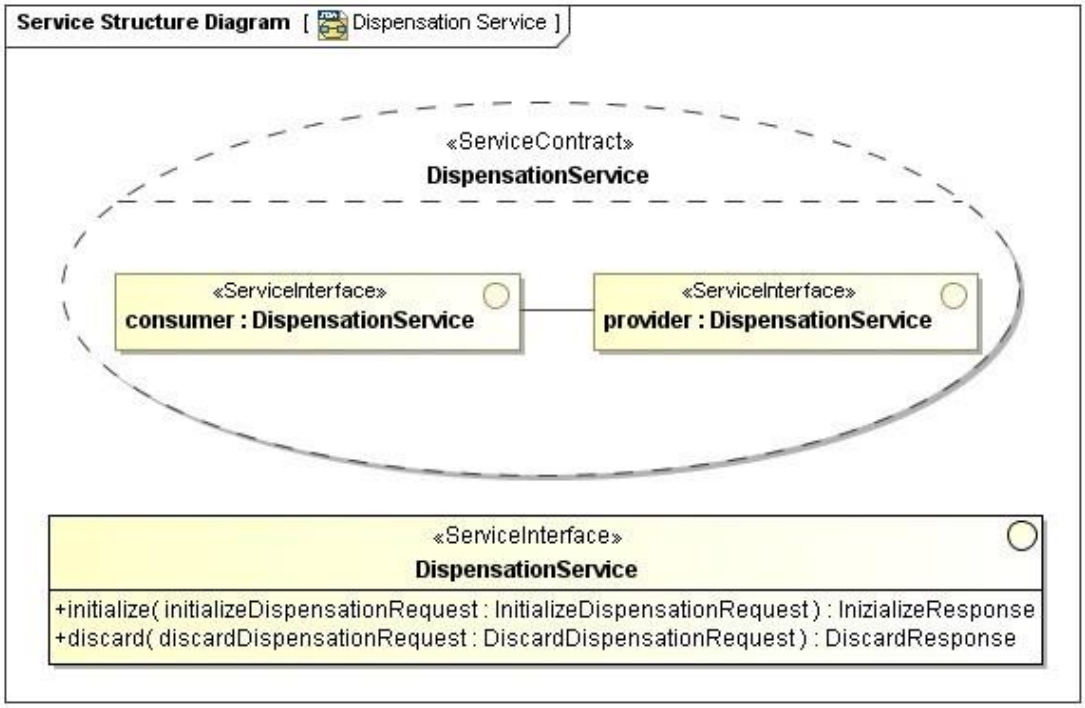


Figure 33: eHealth DSI Dispensation Service Interface

Operation	initialize()	
Description	Notify the patient's country of affiliation on a successful dispensation of an ePrescription	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	initializeDispensationRequest	
	Body	(1) eHealth DSI coded eDispensation documents as defined by [epSOS D3.5.2]. (2) source coded dispensation data The body MUST contain at least one eHealth DSI pivot coded dispen- sation document (1). It MUST contain at least one source coded document (2). There MUST be a 1:1 association among provided source coded documents and epSOS coded eDispensation documents.
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion [ST] eHealth DSI Treatment Relationship Confirmation Assertion [O]
Output Message in successful Case	initializeDispensationResponse	
	Body	(1) Success indicator
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer is able to locate the service provider 2. A secure channel is established between service consumer and provider nodes 3. The service consumer can verify the authenticity of the service provider 4. Service consumer and service provider share a common identifier for the patient 5. The patient has given consent to the use of eHealth DSI 6. The service consumer has previously retrieved the list of the patients available ePrescriptions 7. All available ePrescriptions for the identified patient are accessible for NCP-A and the provided eDispensation data relates to these ePrescriptions 8. The requesting HP has been authenticated in the country of care and the service provider is able to verify the attesting HP identity	

	assertion 9. A treatment relationship exists between the patient and the requesting HP and the attesting assertion can be verified by the service provider 10. The HP is authorised to dispense medication for the patient
Main success scenario	Actions of the eHealth DSI Dispensation Service provider: 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion and TRC assertion 3. Verify that the patient has given consent to eHealth DSI and that the consent is valid 4. Enforce national security policy and (if available) patient privacy policy 5. Verifiy that all dispensation information is provided and that dispensation data is properly coded 6. Retrieve patient's available prescriptions and verify that each dispensation item matches with a prescribed item 7. Process the dispensation information 8. Apply eHealth DSI security measures to the success indicator and send it to the requestor
Fault Conditions	Preconditions for a success scenario are not met
	The requesting HP has insufficient rights to dispense the identified patient's ePrescriptions
	One or more of the provided dispensation items do not relate to available ePrescriptions of the identified patient
	The ePrescription that is referred to by an eDispensation has already been dispensed.
	No consent for ePrescription sharing and dispensing is registered for the identified patient
	The eDispensation data is not provided in all mandatory encodings
	Temporary failure (e. g. authenticity verification cannot be performed due to a PKI failure)
Warning Conditions	eDispensation data is not processed by the patient's country of affiliation

Operation	discard()	
Description	Notify the patient's country of affiliation on an erroneous eDispensation notification, to allow it to roll back any changes made on its internal data that were triggered by the erroneous notification	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	discardDispensationRequest	
	Body	(1) Identifier of the eDispensation document that is to be discarded
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion
Output Message in successful Case	discardDispensationResponse	
	Body	(1) Success indicator
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer is able to locate the service provider 2. A secure channel is established between service consumer and provider nodes 3. The service consumer is able to verify the authenticity of the service provider 4. Service consumer and service provider share a common identifier for the patient 5. The patient has given consent to the use of eHealth DSI 6. The service consumer has previously retrieved the list of the patients available ePrescriptions an dispensed the identified medicine 7. The requesting HP has been authenticated in the country of care and the service provider is able to verify the attesting HP identity assertion	
Main success scenario	Actions of the eHealth DSI Dispensation Service provider: 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion 3. Extract the dispensed item id from the message body and ensure that this item was previously dispensed by the identified HP 4. Enforce national security policy and (if available) patient privacy policy 5. Rewind the dispensation 6. Sign the success notification and send it to the requestor	
Fault Conditions	Preconditions for a success scenario are not met	
	The HP has insufficient rights to process the patient's ePrescription data	
	The HP was not the original dispenser of the identified medication item	
	The identified item had not been dispensed previously	
	Temporary failure (e.g. service provider is temporarily unable to access an internal service)	
Warning Conditions	eDispensation data is not processed by the country of affiliation	
	eDispensations are not rolled back automatically by the country of affiliation	

5.1.6 eHealth DSI OrCD Service

The eHealth DSI OrCD Service provides operation for retrieving an identified patient's available original clinical documents.

Operation	list()	
Description	Obtain the eHealth DSI-encoded, available original clinical documents of the identified patient	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	ListOrCDRequest	
	Body	Identifier of the patient whose available original clinical documents are requested
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion [ST] ehealth DSI Treatment Relationship Confirmation Assertion [O]
Output Message in successful Case	ListOrCDResponse	
	Body	(1) List of available original clinical documents
	Security Token	[PT] X.509 NCP-A service certificate

Precondition of success scenario	11. The service consumer can locate the service provider 12. A secure channel is established between service consumer and service provider nodes 13. The service provider can verify the authenticity of the service consumer 14. service consumer and service provider share a common identifier for the patient 15. The patient has given consent to the use of eHealth DSI 16. All available original clinical documents for the identified patient are accessible for NCP-A 17. The requesting HP has been authenticated in the country of care and the service provider is able to verify the attesting HP identity assertion 18. A treatment relationship exists between the patient and the requesting HP and the attesting assertion can be verified by the service provider 19. the HP is authorised to access the requested data 20. the service consumer can verify the authenticity of the service provider
Main success scenario	Actions of the eHealth DSI OrCD Service provider: 10. Validate the authenticity of the service consumer 11. Verify HP identity assertion and TRC assertion 12. Verify that the patient has given valid consent to eHealth DSI and that the consent applies to the current usage scenario 13. Retrieve patient's available original clinical documents 14. Enforce national security policy and (if available) patient privacy policy 15. Verify authenticity and integrity of available original clinical documents. 16. Apply eHealth DSI protection means to the response message and send it to the requestor
Fault Conditions	Preconditions for a success scenario are not given
	Requestor has insufficient rights to access the patient's original clinical documents.
	No original clinical documents are available for the identified patient
	No consent for original clinical document sharing is registered for the identified patient
	Temporary failure (e. g. authenticity verification cannot be performed due to a PKI failure)
Warning Conditions	Country A allows for data hiding; a respective disclaimer SHOULD be shown to the HP
	More original clinical documents MAY be available but are not accessible

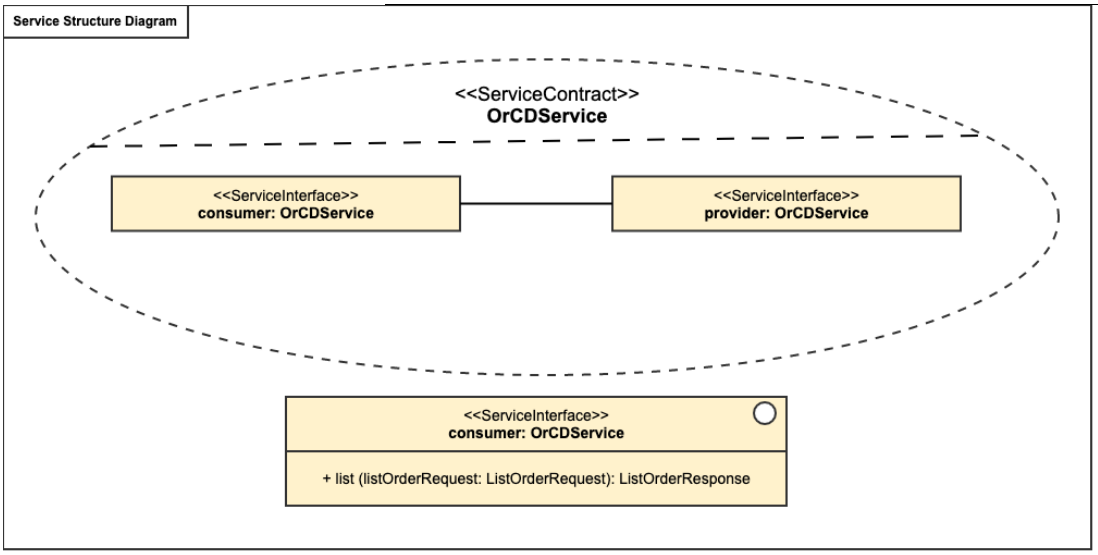


Figure 34 eHealth DSI OrCD Service Interface

5.1.7 eHealth DSI Patient Identification Service (FHIR)

The eHealth DSI Patient Identification Service FHIR implementation provides operation to get a list of patients matching a specific search criteria.

Operation	searchByUrl	
URL	GET $server/Patient?$listOfParameters	
Description	Obtain a patient or list of patients that match the search parameters	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	$listOfParameters	
	Parameters	• Identifier • name • birthdate • gender
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion
Output Message in successful Case	JSON object	
	Body	(1) List of patients that match the search criteria
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	1. The service consumer can locate the service provider 2. A secure channel is established between service consumer and service provider nodes 3. The service provider can verify the authenticity of the service consumer 4. An HP identity assertion has been issued for the requesting HP 5. The service provider can verify the validity of the HP identity assertion 6. the service consumer can verify the authenticity of the service provider 7. The patient has given a consent that authorises NCP-A to disclose his identity	
Main success scenario	1. Actions of the eHealth DSI Patient Identification (FHIR):Validate the authenticity of the service consumer 2. Verify HP identity assertion 3. Verify that the requesting HP is authorised to query for patient information 4. Search for patients that match the provided parameters 5. Get a JSON object with the list of patients that matched the search criteria.	
Fault Conditions	Preconditions for a success scenario are not met	
	Requestor has insufficient rights to access the patient's medical summary	
	No patients match the provided search criteria	
	The patient we were looking for didn't authorize the share of his patient data	

5.1.8 eHealth DSI Document Reference Service (FHIR)

The eHealth Document Reference Service FHIR implementation provides operation to retrieve lab report documents associated with a patient

Operation	searchByUrl	
URL	GET $server/DocumentReference?$listOfParameters	
Description	Obtain a list of lab reports that match the search parameters	
Requestor	Consuming Gateway at NCP-B (service consumer at the country of care)	
Input Message	$listOfParameters	
	Parameters	• patient.identifier (mandatory) • Documenttype (optional) • contenttype (optional) • studytype (optional) • date (optional)
	Security Token	[PT] X.509 NCP-B service certificate [ST] eHealth DSI HP Identity Assertion

Output Message in successful Case	JSON object	
	Body	(1) List of document references that match the search criteria
	Security Token	[PT] X.509 NCP-A service certificate
Precondition of success scenario	8. The service consumer is able to locate the service provider 9. A secure channel is established between service consumer and service provider nodes 10. The service provider is able to verify the authenticity of the service consumer 11. An HP identity assertion has been issued for the requesting HP 12. The service provider is able to verify the validity of the HP identity assertion 13. the service consumer is able to verify the authenticity of the service provider 14. The patient has given a consent that authorises NCP-A to disclose his identity	
Main success scenario	Actions of the eHealth DSI Document Reference Service (FHIR): 1. Validate the authenticity of the service consumer 2. Verify HP identity assertion 3. Verify that the requesting HP is authorised to query for patient information 4. Search for document references that match the provided parameters 5. Get a JSON object with the list of document references that matched the search criteria.	
Fault Conditions	Preconditions for a success scenario are not met	
	Requestor has insufficient rights to access the requested patient medical data.	
	The patient we are looking for does not exist	
	The patient we were looking for didn't authorize the share of his patient data	

5.1.9 Anomaly Detection

eHealth NCP include an Anomaly Detection System. This subcomponent is responsible for analyzing the audit trail and based on a configurable rule to detect possible anomalies (such as excessive requests issued from a HP or a patient).

EHDSI_WARNING_SEC_UNEXPECTED_NUMBER_OF_REQUESTS_FOR_UNIQUE_PATIENT	Too many consecutive requests issued to an NCP-A for a specific patient (threshold of requests is triggered if the event happens in a specified time interval). The count is kept even In case of a request for a non existing patient.
EHDSI_WARNING_SEC_UNEXPECTED_NUMBER_OF_REQUESTS_FOR_UNIQUE_POINT_OF_CARE	Too many consecutive requests issued to an NCP-A from a unique point of care (threshold of requests is triggered if the event happens in a specified time interval)
EHDSI_WARNING_SEC_UNEXPECTED_NUMBER_OF_REQUESTS	Too many requests issued to an NCP-A (threshold of requests is triggered if the event happens in a specified time interval)

The triggering event is persisted in the relative audit operation (XCPD, XCA or XDR) as a warning. The messaging exchange is not impacted. Anomaly detection for general traffic rules (UNEXPECTED_NUMBER_OF_REQUESTS, _FOR_UNIQUE_POINT_OF_CARE) results in audit warnings only and does not block traffic. However, anomaly detection applied to emergency access (EMERGENCY purpose-of-use) from a single HP identity exceeding the country-defined threshold MUST result in a deny response, consistent with Section II §2.5.

5.2 OBJ Audit Component OBJ

This Audit Component is responsible for formatting an EventLog message in an Audit Trail and Node Authentication (ATNA) -compatible way, and pass it securely to the audit repository.

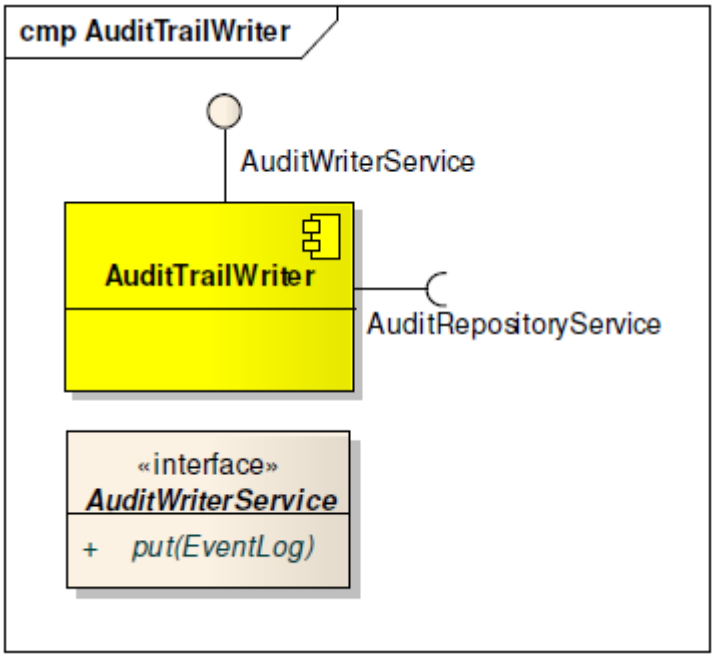


Figure 38: AuditTrailWriter component

5.3 Audit Repository

The AuditRepository component is responsible for storing audit trail entries in an ATNA-compatible way. The content of the audit repository will be analyzed in the national infrastructure. Therefore, presented here the AuditService interface has a very general structure that can be further adjusted by countries according to their specific needs.

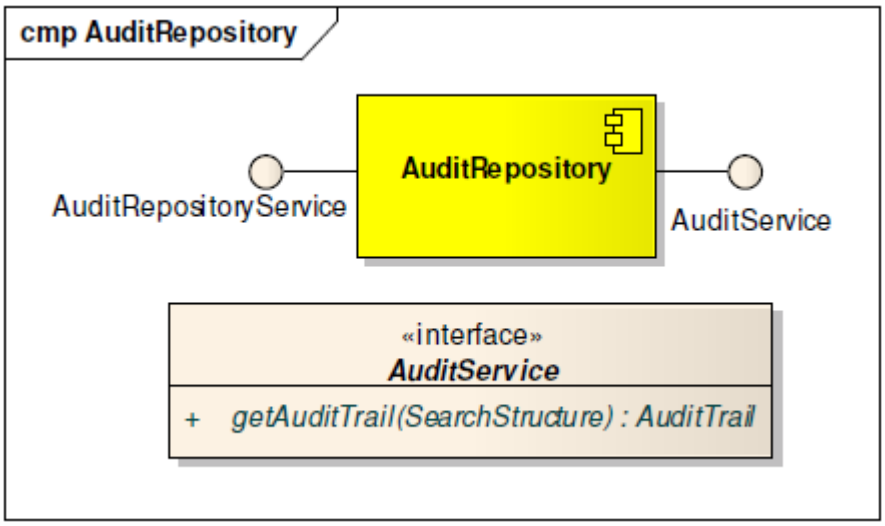


Figure 39: AuditRepository component

getAuditTrail(SearchStructure) : AuditTrail

Textual description of operation	
The operation allows to selectively extract records from the audit trail.	
Input parameters	**SearchStructure** The structure contains search parameters that define which records of the audit trail should be extracted
Output parameters	**AuditTrail** Records of the audit trail that correspond to the parameters defined in SearchStructure

Notes	No identifiable medical information should be contained in the audit repository

5.4 Security Module

The Security module is responsible for creation and verification of digital signatures that are applied to medical documents.

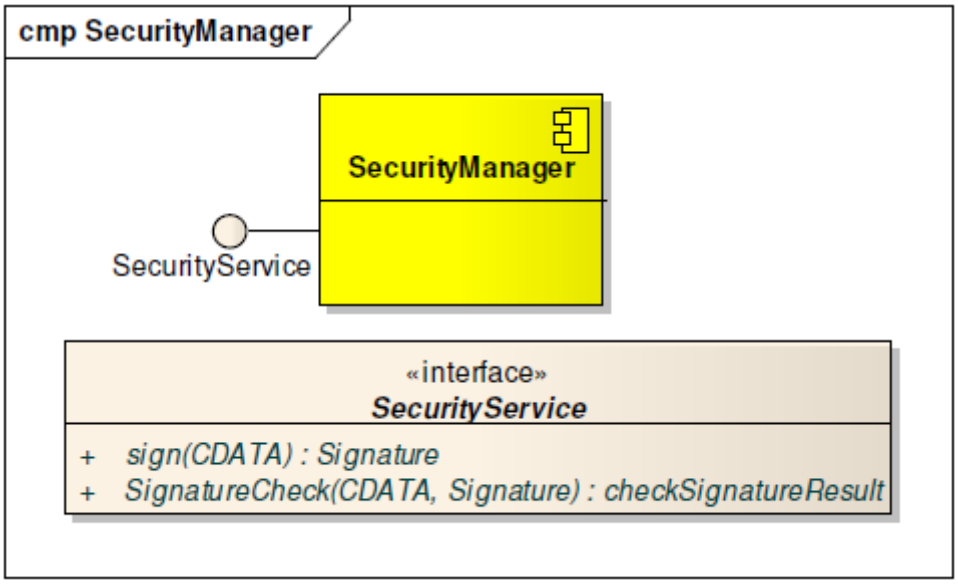


Figure 40: SecurityManager component

sign(CDATA) : Signature

Textual description of operation	
This operation processes a XML DSig signature for a given XML according to the XML DSig standard and the eHealth DSI profilation. Only known documents are signed and before the signature processing a schema validation is done. The documents that are known are configurable.	
Input parameters	**CDATA** XML Document to sign
Output parameters	**Signature**
Notes	In case it is decided that it is enough to provide WS-signatures, this functionality can be discarded

checkSignature(CDATA, Signature) : CheckSignatureResult

Textual description of operation	
The XML DSig signature of a XML document is validated including the validation of the certificate that confirmed the signature.	
Input parameters	**CDATA** XML containing **Signature** Digital signature
Output parameters	**CheckSignatureResult** Encoded in the status information are • Validity of signature • Validity of certificate

Notes	All the known and trusted certificates have to be registered by configuration beforehand. In case it is decided that it is enough to provide WS-signatures, this functionality can be discarded

5.5 Translations and mappings services

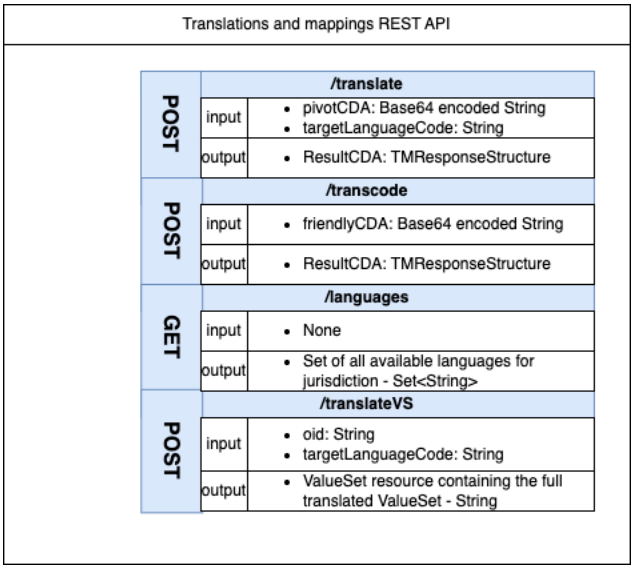


Figure 40: Translations and mappings services

Service responsibilities

- Translating and/or transcoding (if necessary) the original data compliant to eHealth DSI CDA syntax from the national language and possibly from the national code system(s) in the document creator country (in most cases Country A) to the eHealth DSI Reference Terminology. From a functional point of view the translation and transcoding are the same operation for the Transformation Manager from the point of view of the document creator country (in most cases Country A).
- Creating an eHealth DSI unstructured CDA by embedding the original data from document creator presented in the pdf format. Note: the assumption is made that the original data is already presented in a pdf format. The document creator country may have other formats internally; however this data must be presented in a pdf format so that the document consumer country can read it. If, for whatever reasons, the original data needs to be sent, it can be processed in a similar way. However, for the pilot purposes only the pdf operation is defined.
- Retrieval of a list of available translation languages based on a property value corresponding with the AVAILABLE_TRANSLATION_LANGUAGES key in the EHNCP_PROPERTY database.
- Translating the coded data elements from the eHealth DSI Reference Terminology to the national language in document consumer country (in most cases Country B).
- Translation of a full value set into a target language.

1. Transcode

Textual description of operation	
Transformation of national data to eHealth DSI pivot format.	
Input parameters	• **friendlyCDA :** Base64 encoded string of the medical document in its original data format as

	provided from the NationalConnector to this component. [Mandatory]
Output parameters	**TMResponseStructure :** Response structure including the eHealth DSI pivot CDA and the response status structure. The response status structure provides information about the operation results, including possible errors and warning.
Component behaviour	After having received a transcode() request, this component takes the OriginalData (already compliant to eHealthDSI CDA syntax) and using the TSAM capabilities, accomplishes the eventual transcoding of the terms present in the eHealth DSI value sets, while also keeping the original codes and display name. An eHealth DSI pivot document with eHealth DSI coded concepts is therefore produced. The eHealth DSI pivot document shall have a link to the OriginalData. Exceptions: in case of processing error or warning, the responseStatusStructure will be used to convey this information to the calling component with an appropriate error and warning code. A detailed list of the managed exceptions will be provided in the Detail Design Specification document. Each exception condition occurred will be logged (standard and audit), reporting both the exception code and its English description.
Notes	

2. Languages

Textual description of operation	
Operation to retrieve a set of languageCodes (as a combination of an ISO 639-1 language code and an ISO 3166-1 country code, separated by a hyphen) for translations that are shared with a specific jurisdictional domain by querying the AVAILABLE_TRANSLATION_LANGUAGES property in the EHNCP_PROPERTY database. *Exceptions*: in case the AVAILABLE_TRANSLATION_LANGUAGES property is not set.	
Input parameters	
Output parameters	**Set(<String>** a set of all language codes linked with the specific jurisdictional domain.
Notes	

3. Translate

Textual description of operation	
Translation from eHealth DSI pivot data to consumer country language.	
Input parameters	• **pivotCDA: Base64 encoded String of CDA** Document in eHealth DSI pivot format (with eHealth DSI codes) • **targetLanguageCode**: translation target language
Output parameters	• **resultCDA:** Base64 encoded String containing eHealth DSI pivot CDA with translated eHealth DSI

	codes into the consumer country language. The resultCDA is wrapped in a TMResponseStructure, containing also the encountered errors/warnings during the translation process.
Component behaviour	After having received a translate() request, this component starts to process the received CDA in order to extract the eHealth DSI coded concepts. Subsequently, for each coded concept found, it makes use of the TSAM capabilities to obtain the representation of that concept in the target TargetLanguageCode identifier. This information is therefore used by this component to update the displayName attribute of that coded entry. After the completion of this translation phase, an eHealth DSI pivot document with "translated" concepts is obtained. This document is therefore returned to the requesting party. No changes are applied to the document identifiers. Exceptions: in case of processing error or warning, the responseStatusStructure will be used to convey this information to the calling component with an appropriate error and warning code. A detailed list of the managed exceptions will be provided in the Detail Design Specification document. Each exception condition occurred will be logged (standard and audit), reporting both the exception code and its English description.
Notes	

4. TranslateVS

Textual description of operation	
Translation of a full Value set where all the concepts are translated in the passed targetLanguageCode.	
Input parameters	• **oid:** OID of the value set to be translated. • **targetLanguageCode**: translation target language (code) of the target language.
Output parameters	• **valueSet:** Base64 encoded String containing the ValueSet resource containing all the requested translated concepts.
Notes	

Linking documents

The relationship between the eHealth DSI pivot document and the embedded PDF CDA is always inferred via the XFRM relationship with their parent document OriginalData. Optionally, a direct relationship between the eHealth DSI pivot document and the embedded PDF CDA could be created on request as RPLC relationship.

Component behaviour (called operations)

getConceptByCode()

This component issues a getConceptByCode() request in order to know the best matching eHealth DSI Concept, according to the information provided.

Exceptions: in case of returned errors this component shall appropriately use NullFlavors for valorising the "main" coded concept; nevertheless the original code shall be provided through the <translation> element.

getDesignationByConcept()
This component issues getDesignationByConcept() request in order to know the target language eHealth DSI Designation, according to the information provided.

Exceptions: in case of returned errors no actions are required for displayName translation.
This component provides capabilities for coded concepts translation and transcoding if necessary.

5.6 Terminology Services Access Manager

This component provides capabilities for coded concepts translation and transcoding if necessary.

The first term (translation) stands for the capability of associating to an eHealth DSI coded concept the localized concept description or display name: i.e. the translation into the target language of the "concept" conveyed (e.g. code "30001000" EDQM can have the display names "Φύσιγγα", "Ampulka" or "Ampoule", depending on where it is used.).
With the second term (transcoding) we mean the capability of getting the eHealth DSI quasi-synonymous[5] associated to a "local" coded concept.
In both cases, the content needed by the Terminology Services Access Manager is the Terminology Repository. The Terminology Repository is a database representation of the eHealth DSI Reference Terminology.
It must be noted that the eHealth DSI Reference Terminology has as a starting point the eHealth DSI MVC, which in turn is the basis for the eHealth DSI MTC. The mapping activity from the "local" coded concept to the eHealth DSI Value Sets present in the eHealth DSI MVC is out of scope of eHealth DSI and it is the responsibility of the National Linguistic Competence Centers from each country. This mapping is compiled in the eHealth DSI Master Value Sets Translation/Transcoding Catalogue (eHealth DSI MTC). The maintenance of the MTC and its relationship with the MVC are out of scope of the common components design.

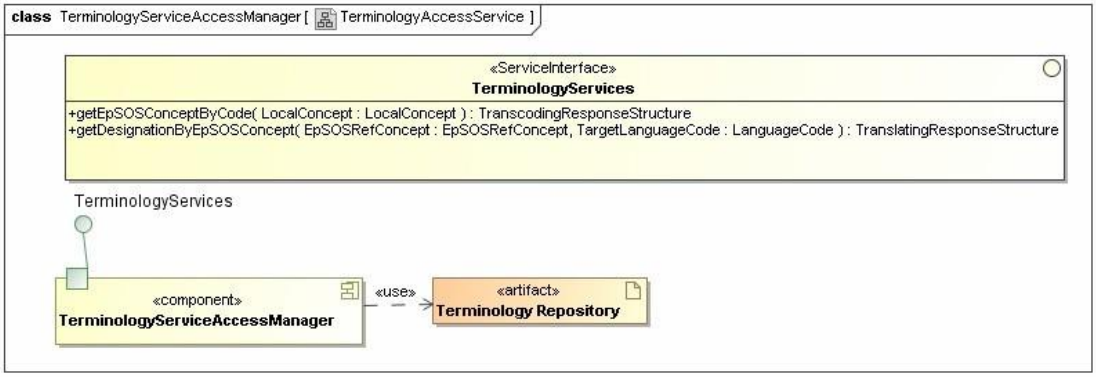


Figure 42: TerminologyServiceAccessManager component

Component responsibilities

1. Translating a given concept designation into the requested target language using the information present in the Terminology Repository.
2. Transcoding a given "local" coded concept into the appropriate eHealth DSI coded concept using the information present in the Terminology Repository.

[5] i.e. a coded concept derived from the appropriate eHealth DSI Value Set semantically equivalent to a given coded concept.

getConceptByCode(LocalConcept) : TranscodingResponseStructure

Textual description of operation	
Transcoding a given "local" coded concept into the appropriate eHealth DSI coded concept using the information present in the Terminology Repository.	
Input parameters	**LocalConcept** structure used to convey the concept derived from the eHealth DSI Original Data. It shall include at least the concept code and the concept code system. Code System Version, Country Code and value set OID - if available - should be provided.
Output parameters	**TranscodingResponseStructure** Response structure including: 1. the eHealth DSI Reference Concept: this means the Concept Code, the English designation, the concept code system (OID), Code System Version, Value Set OID; Value Set Version, 2. the responseStatusStructure, providing information about operation result, including possible errors and warning.
Component behaviour	When this component receives a getConceptByCode() request, it uses all the data extracted from the LocalConcept structure in order to search within the Terminology Repository for the best matching eHealth DSI Concept, according to the local information provided (e.g., if no code system version is indicated, the latest version will be provided). All information retrieved is finally returned to the requesting component. Exceptions: if there is no transcoding or a processing error occurs, the responseStatusStructure will be used to convey this information to the calling component with an appropriated error and warning code. A detailed list of the managed exceptions will be provided in the Detail Design Specification document. Each exception condition occurred will be logged (standard and audit), reporting both the exception code and its English description.
Notes	Makes use of Terminology Repository

getDesignationByConcept(RefConcept; TargetLanguageCode) : TranslatingResponseStructure

Textual description of operation
Translating a given concept designation into the requested target language using the information present in the Terminology Repository

input parameters	**CodeConcept** Structure used to convey the concept derived from the eHealth DSI pivot CDA. It shall include at least the concept code and the concept code system. Code System Version, Country Code and value set OID - if available - should be provided. TargetLanguageCode identifier (code) of the target language.
Output parameters	**TSAMResponseStructure** Response structure including: 1. the target language concept designation; 2. the responseStatusStructure providing information about operation result, including possible errors and warning.
Component behaviour	When this component receives a getDesignationByConcept() request, it uses all the data extracted from the RefConcept structure in order to search within the Terminology Repository for the target language eHealth DSI Designation, according to the local information provided (e.g., if no code system version is indicated, the latest version will be provided). All information retrieved are finally returned to the requesting component. Exceptions: if there is no translation or a processing error occurs, the responseStatusStructure will be used to convey this information to the calling component with an appropriated error and warning code. A detailed list of the managed exceptions will be provided in the Detail Design Specification document. Each exception condition occurred will be logged (standard and audit), reporting both the exception code and its English description.
Notes	Makes use of Terminology Repository

5.6.1 Interaction Uses

This section describes the two NCP "interaction uses" (IU) in which the Semantic Components are involved:

1. The Document Provider Worflow Manager Transformation Manager Interaction Use (Out_WM_TM_IU). This IU is performed at a certain time of the workflow by the Document Provider NCP (usually NCP-A) to create the eHealth DSI pivot CDA and the CDA with the original PDF embedded.
2. The Document Consumer Worflow Manager Transformation Manager Interaction Use (In_WM_TM_IU). This IU is performed at a certain time of the workflow by the Document Consumer NCP (usually NCP-B) to create a translated version of the eHealth DSI pivot CDA into the target language.

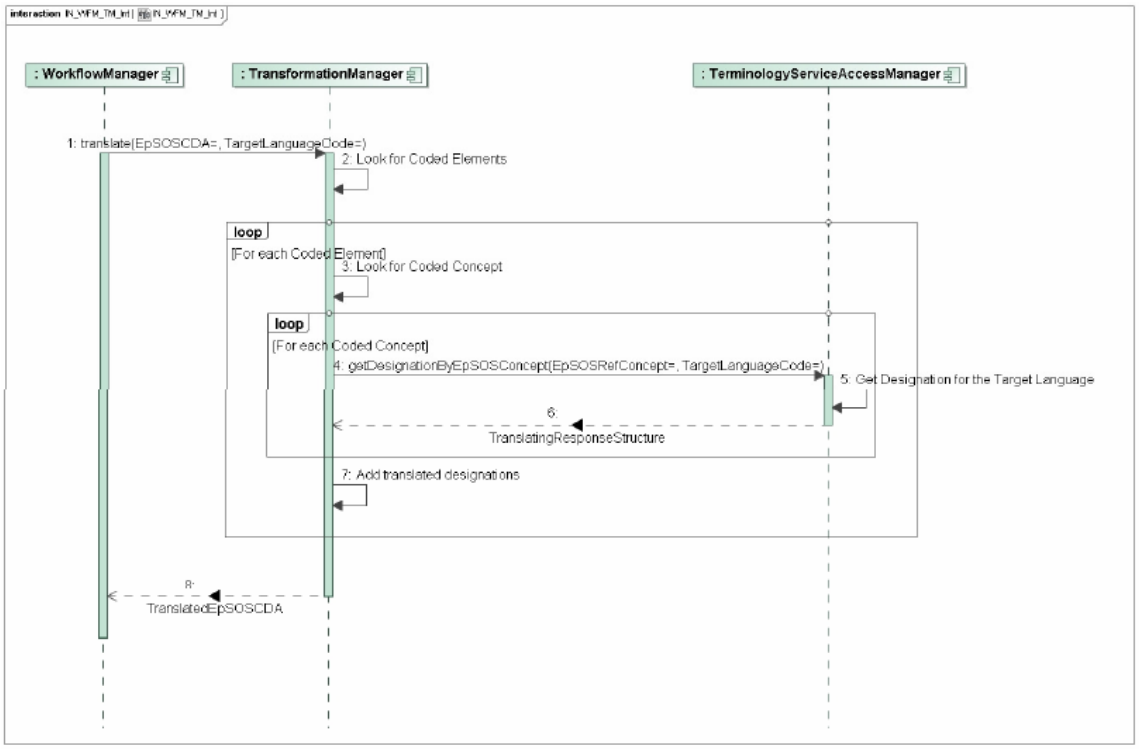


Figure 43: Document Consumer WorkflowManager – TransformationManager Interaction Use (IN_WM_TM_IU)

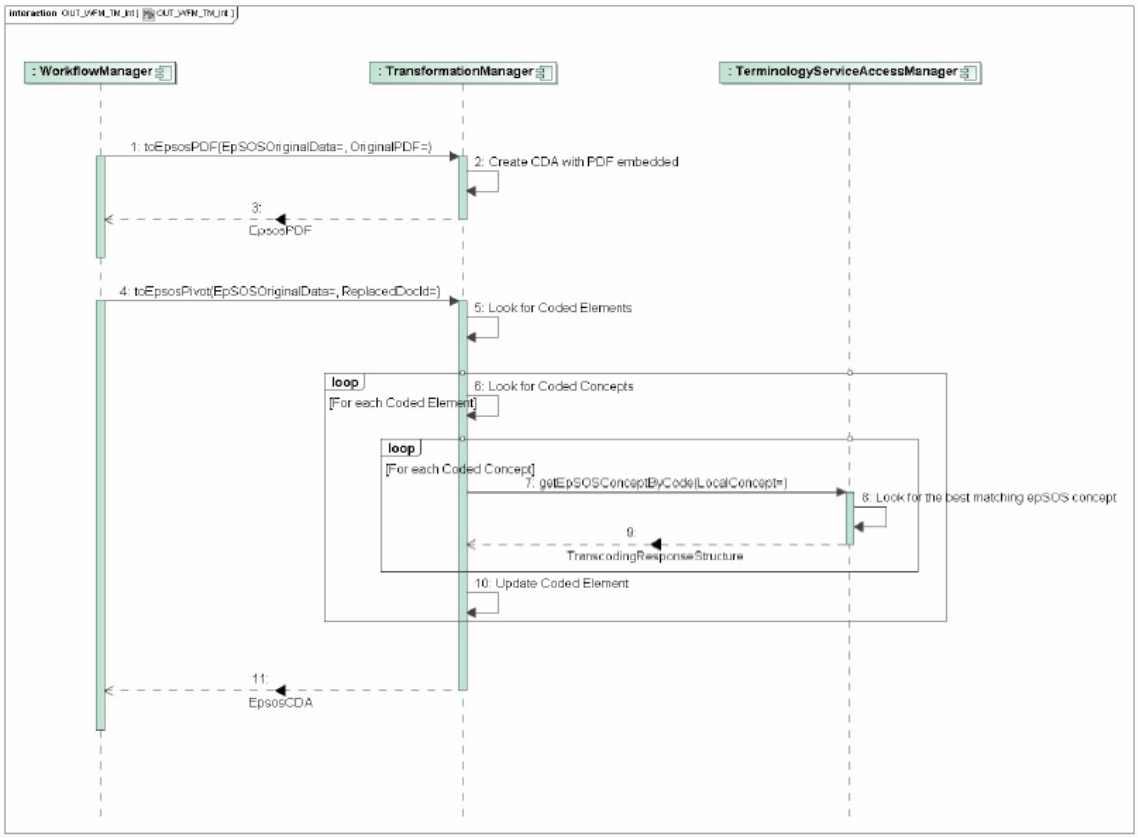


Figure 44: Document Provider WorkflowManager – TransformationManager Interaction Use (OUT_WM_TM_IU)

5.7 NationalConnector

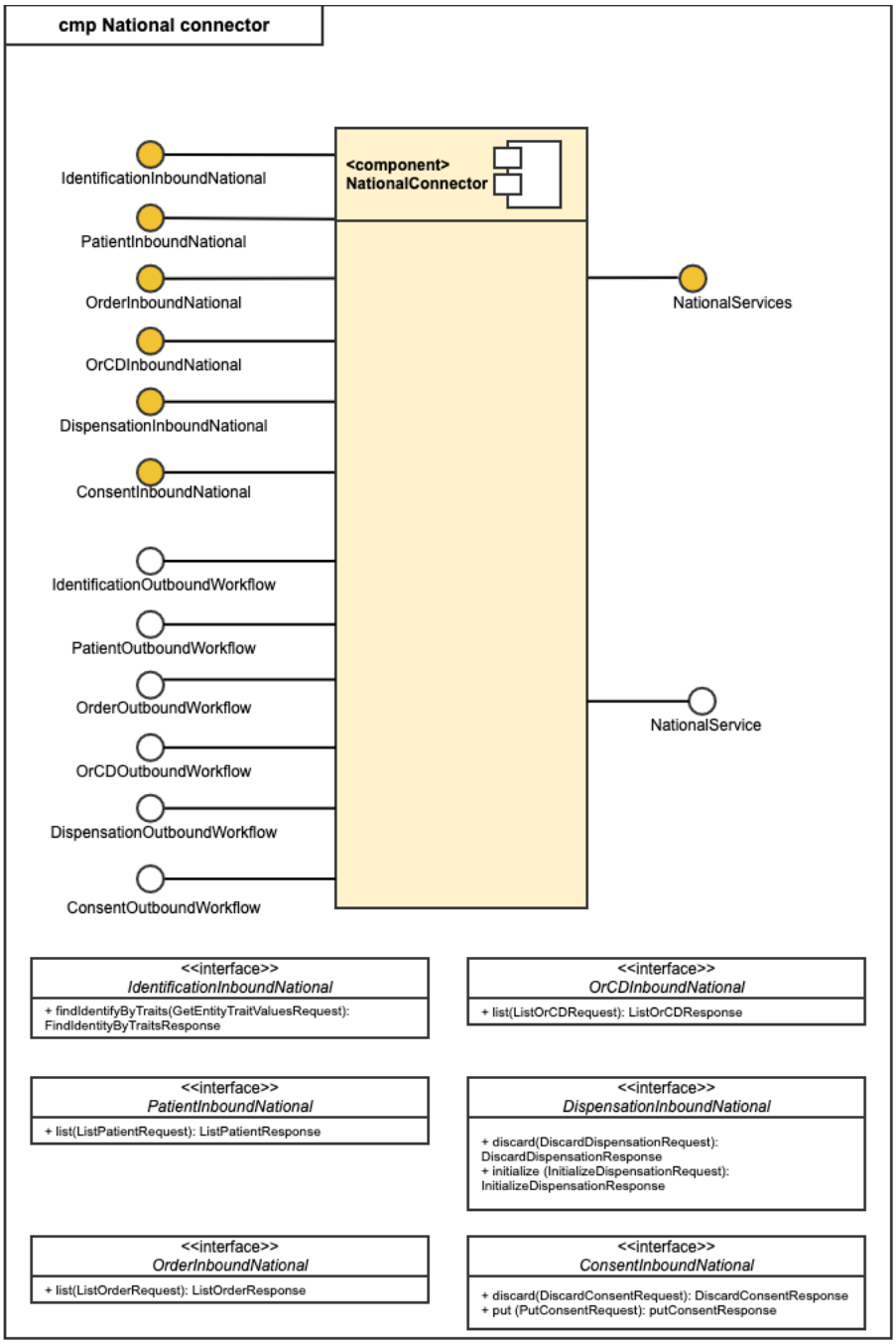


Figure 45: NationalConnector component

The NationalConnector is a component that is developed by countries and is responsible for connecting the NCP to the national infrastructure of the country. In order to ensure that the commonly developed components can interoperate with the NCP-in-the-box, interfaces the NationalConnectors exposed to the WorkflowManager should be kept consistent across all individual implementations.

Interfaces of the NationalConnector exposed to the national infrastructure are defined by a country and therefore are not described in the document.